

Gattmann Algebraic geometry

Yosime Flood

September 2025

1 Chapter 1: Affine Varieties

1.19 Prove that every finite collection of points in $\mathbb{A}^n$ can be written as the zero locus of n polynomials.

Proof. Let $X = \{(a_1^i, \dots, a_n^i) \mid i = 1, \dots, k\} \subseteq \mathbb{A}^n$ be the finite set in question. We will first assume that $a_1^i \neq a_1^j$ for $i \neq j$. The general case will then follow from a suitable change of coordinates. We define our first polynomial f_1 as:

$$f_1 = \prod_{i=1}^k (X_1 - a_1^i) \in k[X_1, \dots, X_n].$$

Notice that a point is a zero of this function if and only if it shares the X_1 coordinate with one of the points in X . Define the remaining $n - 1$ polynomials $f_2, \dots, f_n$ as:

$$f_i = \sum_{j=1}^k \left(\prod_{l \neq j} (X_1 - a_1^l) \right) (X_i - a_i^j).$$

We claim that $X = V(f_1, \dots, f_n)$. Indeed, suppose first that $p \in V(f_1, \dots, f_n)$. Then since p is a zero of f_1 , its X_1 -coordinate agrees with exactly one of the points in our set. In other words, if $p = (p^1, \dots, p^n)$ then $p^1 = a_1^t$ for exactly one $t \in \{1, \dots, k\}$. Since p is also a zero of f_i , we find:

$$\begin{aligned} 0 = f_i(p) &= \sum_{j=1}^k \left(\prod_{l \neq j} (p^1 - a_1^l) \right) (p^i - a_i^j). \\ &= \left(\prod_{l \neq t} (p^1 - a_1^l) \right) (p^i - a_i^t) \end{aligned}$$

Since the left factor is nonzero, the right factor must be zero. In particular, since this holds for all i , we find $p = (a_1^t, \dots, a_n^t)$. From the above term, one easily sees that all the points in X do in fact lay in $V(f_1, \dots, f_n)$.

To finish the proof, it suffices to choose a suitable choice of coordinates¹ such that w.r.t the new coordinate system no two of the points share a common X_1 coordinate. Indeed, in such a coordinate system with coordinate change $T: \mathbb{A}^n \rightarrow \mathbb{A}^n$ (mapping our set of points to a new set $T(X)$ which fulfills the property), we would be able to find polynomials $f_1, \dots, f_n$ as above, whose zero locus is precisely $T(X)$. Then, the zero locus of $f_1 \circ T, \dots, f_n \circ T$, is precisely X . $\square$

1.20 Let X be an affine variety. prove that $A(X)$ is a field if and only if X is a point.

Proof. $A(X)$ is a field if and only if its unique maximal ideal is (0) . Since the ideals of $A(X)$ are in one to one correspondence with the ideals containing $I(X)$, this implies, that $I(X)$ is maximal. By Hilbert's Nullstellensatz, $V(I(X))$ is nonempty, and any point in this set corresponds to a maximal ideal containing $I(X)$. Since $I(X)$ was maximal, and different points correspond to different maximal ideals, $V(I(X)) = X$ contains a unique point. $\square$

1.21 Determine the radical of the ideal

$$\langle x_1^3 - x_2^6, x_1x_2 - x_2^3 \rangle \subseteq \mathbb{C}[x_1, x_2].$$

We claim that the radical is given by

$$\sqrt{\langle x_1^3 - x_2^6, x_1x_2 - x_2^3 \rangle} = \langle x_1 - x_2^2 \rangle.$$

Proof. By Hilbert's Nullstellensatz, it suffices to show that

$$I(V(x_1^3 - x_2^6, x_1x_2 - x_2^3)) = \langle x_1 - x_2^2 \rangle.$$

Suppose we have a point $p \in V(x_1^3 - x_2^6, x_1x_2 - x_2^3)$, Then its coordinates satisfy:

$$\begin{cases} x_1^3 - x_2^6 &= 0 \\ x_1x_2 - x_2^3 &= 0. \end{cases}$$

Plugging in the second condition into the first yields $x_1^2(x_1 - x_2^2) = 0$ In particular, one of the factors is zero. The defining equations above yield that if x_1 is zero then so is x_2 and thus in either case $x_1 - x_2^2 = 0$. This proves that

$$\langle x_1 - x_2^2 \rangle \subseteq I(V(x_1^3 - x_2^6, x_1x_2 - x_2^3)).$$

The reverse inclusion follows by observing the inclusion

$$V(\langle x_1 - x_2^2 \rangle) \subseteq V(x_1^3 - x_2^6, x_1x_2 - x_2^3).$$

¹The existence of this coordinate system is intuitively obvious (almost all systems work). However, it is slightly tedious to actually prove it.

If we apply I to this relation, we find

$$\sqrt{\langle x_1 - x_2^2 \rangle} \supseteq I(V(x_1^3 - x_2^6, x_1x_2 - x_2^3)).$$

Finally since x_1 is not a square in the ring $\mathbb{C}[x_1]$, the polynomial $x_1 - x_2^2 \in \mathbb{C}[x_1][x_2] = \mathbb{C}[x_1, x_2]$ is irreducible and thus prime. In particular $\langle x_1 - x_2^2 \rangle$ is a prime ideal and hence radical. $\square$

- 1.22 Let $X \subset \mathbb{A}^3$ be the union of the three coordinate axes. Compute generators for the ideal $I(X)$, and show that $I(X)$ cannot be generated by fewer than three elements. Let $k[X_1, X_2, X_3]$ be the coordinate ring of $\mathbb{A}^3$.

Proof. We claim that

$$I(X) = \langle X_1X_2, X_1X_3, X_2X_3 \rangle.$$

Indeed a point P lies in X if and only if two of its coordinates are coordinates equal to zero. Thus;

$$V(X_1X_2, X_1X_3, X_2X_3) = X$$

and so

$$I(V(X_1X_2, X_1X_3, X_2X_3)) = I(X).$$

Therefore, we need only to show that $\langle X_1X_2, X_1X_3, X_2X_3 \rangle$ is radical. Indeed, a polynomial belongs to this ideal if and only if it only contains no pure powers of X_1, X_2, X_3 . For any polynomial p that does contain a pure power of one of the variables, p^n will also contain a pure power in the same variable and hence never will belong to $\langle X_1X_2, X_1X_3, X_2X_3 \rangle$. To complete the task we need to show that $\langle X_1X_2, X_1X_3, X_2X_3 \rangle$ is not generated by two elements. Indeed, this follows from the fact that $I(X) \cap k[X_1, X_2, X_3]_{\leq 2}$ is a three dimensional k vector space. If we had two generators of $I(X)$, then their quadratic terms would form a generating set of the above k vector space which is not possible. $\square$

- 1.23 Let $Y \subset \mathbb{A}^n$ be an affine variety, and denote by

$$\pi : K[x_1, \dots, x_n] \rightarrow K[x_1, \dots, x_n]/I(Y) = A(Y)$$

the quotient map.

- (a) Show that $V_Y(J) = V(\pi^{-1}(J))$ for every ideal $J \subset A(Y)$.
- (b) Show that $\pi^{-1}(I_Y(X)) = I(X)$ for every affine subvariety X of Y .
- (c) Use (a) and (b) to deduce the (interesting part of the) relative Nullstellensatz

$$I_Y(V_Y(J)) \subset \sqrt{J}$$

for every ideal $J \subset A(Y)$ from the corresponding absolute statement

$$I(V(J)) \subset \sqrt{J}$$

for every ideal $J \subset K[x_1, \dots, x_n]$ in Proposition 1.10. In particular, conclude that there is an inclusion-reversing bijection between affine subvarieties of Y and radical ideals in $A(Y)$.

- Proof.* (a) $p \in V_Y(J)$ holds if and only if $p \in Y$ and all functions in J vanish on p . This in turn holds if and only if all functions in $\pi^{-1}(J)$ vanish on p .
- (b) If $f \in \pi^{-1}(I_Y(X))$, then for any point in X we have $f(x) = \pi(f)(p) = 0$ and so $f \in I(X)$. Conversely, if $f \in I(X)$, and $p \in X$, then $\pi(f)(p) = f(p) = 0$ and so $\pi(f) \in I_Y(X)$.
- (c) Suppose $f \in I(V_Y(J))$ and $\tilde{f}$ is a π preimage of f . By (b), $\tilde{f}$ is a function which vanishes on $V_Y(J)$, which in turn, by (a), is means that $\tilde{f}$ vanishes on $V(\pi^{-1}(J))$. By Hilbert's Nullstellensatz, there exists a $n \in \mathbb{N}$ such that $\tilde{f}^n \in \pi^{-1}(J)$. Applying π to this we obtain $f^n \in \sqrt{J}$.

□

2 Chapter 2: The Zariski Topology

2.17 . Find the irreducible components of the affine variety

$$V(x_1 - x_2x_3, x_1x_3 - x_2^2) \subset \mathbb{A}_{\mathbb{C}}^3.$$

We claim that the irreducible decomposition of $V(x_1 - x_2x_3, x_1x_3 - x_2^2)$ is given by

$$V(x_1 - x_2x_3, x_1x_3 - x_2^2) = V(x_1, x_2) \cup V(x_1 - x_3^3, x_2 - x_3^2)$$

Proof. First note that the two above sets are indeed contained in

$$X := V(x_1 - x_2x_3, x_1x_3 - x_2^2)$$

as can be easily verified by a direct check of equations. We will now show the reverse inclusion. Suppose $p \in V(x_1 - x_2x_3, x_1x_3 - x_2^2)$, plugging in the first condition into the second, we find that p 's coordinates satisfy

$$0 = x_2x_3^2 - x_2^2 = x_2(x_3^2 - x_2).$$

Thus either $x_2 = 0$ or $x_3^2 = x_2$. In the first case, the defining equations of X dictate $x_1 = 0$ and so $p \in V(x_1, x_2)$. Let us now assume $x_2 \neq 0$. Then, using the fact that $x_3^2 = x_2$, the defining equations of X become:

$$\begin{cases} x_1 - x_3^3 = 0 \\ x_1x_3 - x_3^4 = 0 \end{cases}$$

any point which satisfies the first equation satisfies the second as well and so we find that $p \in V(x_1 - x_3^3, x_2 - x_3^2)$. To complete the proof, we need to show that the two above components are irreducible (note that they are clearly not contained in one another and both do not equal all of X). This is equivalent to checking that their respective ideals are prime. Denote by $X_1 := V(x_1, x_2)$ and $X_2 := V(x_1 - x_3, x_2 - x_3^2)$. For X_1 we have $I(X_1) = (x_1, x_2)$ which is prime, since $\mathbb{C}[x_1, x_2, x_3]/I(X_1) \simeq \mathbb{C}[x_3]$ which is an integral domain. For X_2 we have $\mathbb{C}[x_1, x_2, x_3]/(x_1 - x_3, x_2 - x_3^2) \simeq \mathbb{C}[t, t^{1/2}, t^{1/3}]$ which proves that $(x_1 - x_3, x_2 - x_3^2)$ is prime and (thus also equal to $I(X_2)$). $\square$

2.18 Let $X \subset \mathbb{A}^n$ be an arbitrary subset. Prove that

$$V(I(X)) = \overline{X}.$$

Proof. Let $C \supseteq X$ be any closed set. Applying $V \circ I$, we find $V(I(X)) \subseteq V(I(C))$. Since C is closed we have $C = V(I(C))$. Thus $V(I(X))$ is the minimal closed set containing X , hence its closure. $\square$

2.19 Let X be a topological space.

- (a) If X is Noetherian show that we can write it as a finite disjoint union $X = X_1 \cup \dots \cup X_r$ of nonempty connected closed subsets of X , and that this decomposition is unique up to permutation. We call $X_1, \dots, X_r$ the connected components of X .
- (b) Let $X_1, \dots, X_r$ be any subsets of X with $X = X_1 \cup \dots \cup X_r$. If all $X_1, \dots, X_r$ are Noetherian, prove that X is Noetherian as well.

Proof. (a) Suppose X is a Noetherian space for which the statement is false. Then X itself is not connected and so X may be written as a disjoint union of closed sets $X_1 \cup Y_1$. Then either X_1 or Y_1 can also not be written as a finite disjoint union of connected sets, since such decompositions would yield a decomposition of X . without loss of generality suppose X_1 cannot be written as such, (or else rename). Once again X_1 cannot be connected so it admits a disjoint decomposition $X_1 = X_2 \cup Y_2$ of connected sets. Proceeding iteratively, we obtain a strictly decreasing chain of closed sets:

$$X \supset X_1 \supset X_2 \supset \dots$$

which is not possible in a Noetherian space. It remains to prove uniqueness. Suppose we have two decompositions:

$$X = X_1 \cup \dots \cup X_r = X'_1 \cup \dots \cup X'_s.$$

This yields a decomposition

$$X_1 = \cup_{i=1}^s (X'_i \cap X_1).$$

Since X_1 is itself connected we have $X_1 = X'_k \cap X_1$ for exactly one of the $k \in \{1, \dots, s\}$. This amounts to $X_1 \subseteq X'_k$. Using the same reasoning for X'_k , we find a $j \in \{1, \dots, r\}$ with $X'_k \subseteq X_j$. However, since $X_i \cap X_j = \emptyset$ for $i \neq j$, we find that $X'_k \subseteq X_1$ and so in total $X'_k = X_1$. Removing both of these sets from the above unions, the result then follows by induction.

- (b) Suppose $C : C_1 \supseteq C_2 \supseteq \dots$ is a descending chain of closed sets in X . We will show that it terminates. C gives rise to r chains C_i in the respective spaces X_i via:

$$C_i : C_1 \cap X_i \supseteq C_2 \cap X_i \supseteq \dots$$

These are descending chains of closed sets in noetherian spaces and hence they all terminate. Choosing a large enough $n \in \mathbb{N}$ we find that that for all $k \geq n$ and for all $i \in \{1, \dots, r\}$ we have

$$C_n \cap X_i = C_k \cap X_i.$$

If we take the union over all i , we find that for all $k \geq n$

$$C_n \cap X_i = C_k \cap X_i,$$

which means C terminates.

□

2.20 Let A be a subset of a topological space X . Prove:

- (a) If $Y \subset A$ is closed in the subspace topology of A then $\overline{Y} \cap A = Y$.
(b) A is irreducible if and only if $\overline{A}$ is irreducible.

Proof. (a) Clearly $Y \subseteq \overline{Y}$ and by assumption $Y \subseteq A$, thus $Y \subseteq A \cap \overline{Y}$. Conversely, since Y is closed in the subspace topology we may write $Y = C \cap A$ for a closed set $C \subseteq X$. Then we find $\overline{Y} \subseteq C$. Intersecting this inclusion with A , we obtain the reverse inclusion as well: $\overline{Y} \cap A \subseteq C \cap A = Y$.

- (b) Suppose first that A is irreducible and $\overline{A} = C_1 \cup C_2$ for two closed sets in $\overline{A}$ (hence also closed in X since $\overline{A}$ is closed). Intersecting this with A we get

$$A = (C_1 \cap A) \cup (C_2 \cap A).$$

Therefore, since A is irreducible, we may assume $A = (C_1 \cap A)$. Taking the closures (in X) we observe

$$\overline{A} \subseteq C_1$$

and thus $\overline{A} = C_1$. Since C_1 and C_2 were arbitrary, $\overline{A}$ is irreducible.

Now suppose that $\overline{A}$ is irreducible and $A = C_1 \cup C_2$ for closed sets (in the subspace topology of A). Taking closures we find:

$$\overline{A} = \overline{C_1 \cup C_2} = \overline{C_1} \cup \overline{C_2},$$

where the second equality can be seen by observing that any closed set containing C_1 and C_2 contains $\overline{C_1}$ and $\overline{C_2}$. Since $\overline{A}$ is irreducible, we may assume $\overline{A} = \overline{C_1}$. Intersecting this equality with A , and using part (a), we find $A = C_1$. □

2.21 Let $\{U_i : i \in I\}$ be an open cover of a topological space X , and assume that $U_i \cap U_j \neq \emptyset$ for all $i, j \in I$. Show:

- (a) If each U_i is connected then X is connected.
- (b) If each U_i is irreducible then X is irreducible.

Proof. (a) Suppose $X = X_1 \cup X_2$ for two disjoint open sets X_1, X_2 . Let U_i be any one of the open sets from our cover. The above decomposition yields $U_i = (X_1 \cap U_i) \cup (X_2 \cap U_i)$. Since U_i is connected, and the above sets are disjoint, we may assume wlog, that $U_i = (X_1 \cap U_i)$. Now let U_j with $j \in I$ be any other open set appearing in our cover. For the same reasons as above U_j is contained in either X_1 or X_2 . In the latter case, we would have $U_i \cap U_j = \emptyset$ which contradicts our assumptions on the cover. Thus, all U_j are in X_1 and $X = X_1$.

- (b) We will show that if V_1, V_2 are two open sets with $V_1 \cap V_2 = \emptyset$ then either V_1 or V_2 is empty. Indeed, if neither are empty, we can find open sets U_1, U_2 from our open cover with $U_1 \cap V_1 \neq \emptyset$ and $U_2 \cap V_2 \neq \emptyset$. However we also have

$$\begin{aligned} (U_1 \cap V_1) \cap (U_1 \cap V_2) &= \emptyset \\ (U_2 \cap V_1) \cap (U_2 \cap V_2) &= \emptyset. \end{aligned}$$

The Irreducibility of U_1 and U_2 thus imply $U_1 \cap V_2 = \emptyset$ and $U_2 \cap V_1 = \emptyset$. This however implies that the two sets $U_1 \cap V_1$ and $U_1 \cap U_2$ are two nonempty disjoint open sets of U_1 which contradicts the irreducibility of U_1 . □

2.22 Let $f : X \rightarrow Y$ be a continuous map of topological spaces. Prove:

- (a) If X is connected then so is $f(X)$.
- (b) If X is irreducible then so is $f(X)$.

Proof. (a) If $f(X) = U_1 \cup U_2$ for disjoint two open sets U_1, U_2 , then $X = f^{-1}(U_1) \cup f^{-1}(U_2)$ is a partition of X into open sets. Thus one of them is empty, and therefore one if the images is empty.

(b) Same as (a) but using that the preimage of closed sets are closed. $\square$

2.23 Recall that for two ideals J_1 and J_2 in a ring R the *ideal quotient* is defined by

$$J_1 : J_2 = \{ f \in R : fJ_2 \subset J_1 \}.$$

Show that ideal quotients correspond to differences of varieties in the following sense: if X is an affine variety and...

(a) Y_1 and Y_2 are subvarieties of X , then

$$I(\overline{Y_1 \setminus Y_2}) = I(Y_1) : I(Y_2) \quad \text{in } A(X);$$

(b) J_1 and J_2 are radical ideals in $A(X)$, then

$$\overline{V(J_1) \setminus V(J_2)} = V(J_1 : J_2).$$

Proof. (a) Suppose f is a function which vanishes on the closure of $Y_1 \setminus Y_2$ and g is any function which vanishes on Y_2 . Then $f \cdot g$ vanishes on all the points of Y_1 since any point lies either in $Y_1 \setminus Y_2$ or in Y_2 . Conversely suppose that f is a function in $I(Y_1) : I(Y_2)$ and let $p \in Y_1 \setminus Y_2$. Then we may choose a function $g \in I(Y_2)$ which does not vanish at p due to the inclusion reversing bijection of varieties and ideals of the form $I(W)$. Then we find that $f(p) \cdot g(p) = 0$ which implies $f(p) = 0$. Thus, $V(f)$ is a closed set containing $Y_1 \setminus Y_2$, which implies $V(f) \supseteq \overline{Y_1 \setminus Y_2}$. In other words, $f \in I(\overline{Y_1 \setminus Y_2})$.

(b) Using the fact that J_1, J_2 are radical, By (a) we now have:

$$\begin{aligned} V(J_1 : J_2) &= V(I(V(J_1) : I(V(J_2))) \\ &= V(I(\overline{V(J_1) \setminus V(J_2)})) \\ &= \overline{V(J_1) \setminus V(J_2)} \end{aligned}$$

$\square$

2.24 Let $X \subset \mathbb{A}^n$ and $Y \subset \mathbb{A}^m$ be irreducible affine varieties. Prove that their product $X \times Y \subset \mathbb{A}^{n+m}$ is irreducible as well.

Proof. Suppose $X \times Y = Z_1 \cup Z_2$ were reducible. for a fixed $y \in Y$ consider the embedding

$$\varphi_y : X \rightarrow X \times Y, \quad x \mapsto (x, y).$$

This map is clearly Zariski continuous, since the φ_y -pullback of a polynomial $f \in A(X \times Y)$ is given of $f(\cdot, y)$ which is a polynomial on X . In particular, the the set

$$\varphi_y^{-1}(Z_1) = \{x \in X \mid (x, y) \in Z_1\}$$

is a closed set. Taking the intersection of these sets over all $y \in Y$, we obtain that the set

$$\{x \in X \mid \{x\} \times Y \subseteq Z_1\} =: C_1$$

is a closed set in X . In the exact same way the set

$$\{x \in X \mid \{x\} \times Y \subseteq Z_2\} =: C_2$$

is also closed. We claim that $X = C_1 \cup C_2$. Indeed, suppose $x \in X$. This x gives rise to a union:

$$Y = \{y \in Y \mid (x, y) \in Z_1\} \cup \{y \in Y \mid (x, y) \in Z_2\}$$

where both sets are closed for the same reasons above. The irreducibility of Y implies that Y equal to one of the two sets and thus x is either in C_1 or in C_2 . Since now $X = C_1 \cup C_2$ we find that $X = C_1$ or $X = C_2$ which implies $X \times Y = Z_1$ or $X \times Y = Z_2$. $\square$

2.30 Let A be an arbitrary subsets of a topological space, prove $\dim A \leq \dim X$

Proof. If $\dim X = \infty$ there is nothing to prove, thus assume $\dim X = n < \infty$. Furthermore, if $A \subseteq X$ is a closed set, then the claim is clear since any chain of irreducible closed sets in A is also one in X . For the general case, suppose we have an increasing chain of closed irreducible sets in A

$$A: \emptyset \neq A_0 \subsetneq A_1 \subsetneq \cdots \subsetneq A_{n'} \subseteq A.$$

By problem 2.20 this leads to a chain of closed irreducible sets in X given by:

$$\emptyset \neq \overline{A_0} \subsetneq \overline{A_1} \subsetneq \cdots \subsetneq \overline{A_{n'}} \subseteq \overline{A}.$$

(note that the strict containment remains strict by part (a) of 2.20). Thus, $n' \leq n$. Since the chain A was arbitrary, the result follows. $\square$

2.33 Let X be the set of all 2×3 matrices over a field K that have rank at most 1. Considered as a subset of $\mathbb{A}^6 = \text{Mat}(2 \times 3, K)$, show that X is an irreducible affine variety and find its dimension.

Proof. Let us first argue that X is affine. This can be seen by observing that a matrix in $\text{Mat}(2 \times 3, K)$ has rank $r \leq 1$ if and only if the determinant of all of its 2×2 -minors vanish. These are polynomial equations in the entries of the matrix and thus X is defined as the zero locus of a family of equations. Next let us argue irreducibility. Any matrix $A \in X$ may be written as a product of the form:

$$A = vw^T$$

for vectors $v \in K^2$ and $w \in K^3$. In particular, we receive a surjective map

$$\varphi: \mathbb{A}^6 \simeq \mathbb{A}^2 \times \mathbb{A}^3 \rightarrow X, (v, w) \mapsto vw^T.$$

Since this map is polynomial, it is Zariski-continuous as well. Thus, by problem 2.22 X is irreducible. By the next problem, to calculate the dimension of X , it suffices to calculate the dimension of a dense subset. Let us write the matrices $A \in X$ in the following coordinates:

$$A = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \end{bmatrix}.$$

Consider the open set $U := \{x_{1,1} \neq 0\}$. Every matrix in this subset can be uniquely be written as:

$$\lambda \begin{bmatrix} 1 & a_1 & a_2 \\ a_3 & a_1 a_3 & a_2 a_3 \end{bmatrix}$$

for a unique $\lambda \in K^*$ and unique $a_1, a_2, a_3 \in K$. Thus we get a bijection of the form:

$$\Phi: \mathbb{A}^4 \setminus \{X_1 = 0\} \rightarrow U, (X_1, X_2, X_3, X_4) \mapsto X_1 \begin{bmatrix} 1 & X_2 & X_3 \\ X_4 & X_2 X_4 & X_3 X_4 \end{bmatrix}$$

This map is clearly a zariski-continuous bijection (infact it is a homeomorphism, but we do not need this fact here). Thus, any strict chain of irreducible closed subsets of U can be pulled back to a strict chain of irreducible closed subsets of $\mathbb{A}^4 \setminus \{X_1 = 0\}$. This set however, is dense in $\mathbb{A}^4$ and thus has dimension 4. From this it follows, that $\dim U \leq 4$. On the other hand we may construct a strict chain of irreducible closed subsets in U of length four as shown below:

$$\emptyset \neq \{x_{1,1} \neq 0, x_{1,2}, x_{1,3}, x_{2,1}\} \subsetneq \{x_{1,2}, x_{1,3}, x_{2,1}\} \subsetneq \{x_{1,3}, x_{2,1}\} \subsetneq \{x_{2,1}\} \subsetneq U$$

where the above sets are meant to be viewed in U not in X (i.e we take the intersection with U). In total, we have proven that $\dim U = 4$. To conclude the proof, we simply observe that U is dense in X (as U comes from a dense subset of $\mathbb{A}^6$) and thus the dimensionality result follows from the the next problem. $\square$

2.34 Let X be a topological space. Prove:

(a) If $\{U_i : i \in I\}$ is an open cover of X , then

$$\dim X = \sup\{\dim U_i : i \in I\}.$$

(b) If X is an irreducible affine variety and $U \subset X$ is a non-empty open subset, then

$$\dim X = \dim U.$$

Does this statement hold more generally for any irreducible topological space?

Proof. (a) Suppose $\dim X \geq n$, i.e. there exists a chain of closed irreducible sets in X of the form:

$$\emptyset \neq X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n \subseteq X.$$

Since the U_i cover X , there is a open set U from this cover such that $X_0 \cap U \neq \emptyset$. Furthermore, the set $X_1 \cap U$ is also irreducible for if it had a reduction; say $X_0 \cap U = (C_1 \cap U) \cup (C_2 \cap U)$ for closed sets in C_i in X , this would lead to a reduction of X :

$$X = (U^c \cap X_n) \cup (C_1 \cap X) \cup (C_2 \cap X).$$

Which in turn implies $X \subseteq C_1$ or $X \subseteq C_2$ which contradicts the reduction of $X_0 \cap U$ that we assumed. For the same reasons $X_i \cap U$ are also all irreducible sets of U . Furthermore, we claim that the containments $X_i \cap U \subseteq X_{i+1} \cap U$ are also strict. If this were not the case, we would again have a reduction of X_{i+1} as:

$$X_{i+1} = (X_{i+1} \cap U^c) \cup X_i$$

where the first set is not all of X_{i+1} by choice of U and so $X_i = X_{i+1}$ which is a contradiction. Thus, we have shown $\dim U \geq n$. For the reverse inequality, 2.30 showed that $\dim U_i \leq \dim X$ for all $i \in I$ and so this inequality also holds after taking the supremum over all $i \in I$:

$$\sup\{\dim U_i : i \in I\} \leq \dim X.$$

- (b) The claim follows by induction over $n := \dim X$. If $n = 0$, X is a point and $U = X$ is the only nonempty open subset of X . Suppose now that, $n > 0$ and that $U \subseteq X$ is any nonempty open subset of X . If $U = X$ the claim is clear, and so let us assume that U is a proper subset of X . Then, due to the relative version of Hilbert's Nullstellensatz there exists a nonzero function $f \in A(X)$ such that $V(f) \cap U \neq \emptyset$. since $A(X)$ is an integral domain, f is not a zero divisor and therefore Krull's principle ideal theorem, ensures that each irreducible component of $V(f)$ has dimension $n-1$. One of these components must intersect U , let us denote it by C . By induction, the set $U \cap C \subseteq C$ has dimension $n-1$, which implies that U has dimension equal to n . This completes the proof for the case when X is an irreducible affine variety. If X is only assumed to be an irreducible topological space, the claim is false in general. Consider the space $X = \mathbb{N}$ with the topology τ given by

$$\tau = \{\emptyset, \{1\}, \mathbb{N}\}.$$

One is easily convinced that this is indeed an irreducible topological space. However $U = \{1\}$ has dimension zero, whilst $\mathbb{N}$ has dimension equal to one as can be seen by the chain:

$$\mathbb{N} \setminus \{1\} \subsetneq \mathbb{N}.$$

□

- 2.35 Let X be an affine variety with irreducible decomposition $X = X_1 \cup \cdots \cup X_r$, and let $a \in X$. Show that the local dimension $\text{codim}_X\{a\}$ of X at a is given by

$$\text{codim}_X\{a\} = \max\{\dim X_i : a \in X_i\}.$$

Proof. Suppose first that $\text{codim}_X\{a\} \geq n$. In this case, there exists a chain of closed irreducible subsets of X which has the form:

$$a \in C_0 \subsetneq \cdots \subsetneq C_n \subseteq X.$$

Since C_n is irreducible, it is contained in one of the X_i and thus, $\dim X_i \geq n$ as well. Thus we have shown

$$\text{codim}_X\{a\} \leq \max\{\dim X_i : a \in X_i\}.$$

Conversely suppose there is a component X_i of X which contains a and has dimension $\dim X_i \geq n$. By Hilbert's Nullstellensatz, we may choose a nonzero function $f \in A(X_i)$ which vanishes at a . Krull's Principle function guarantees, that the irreducible components of $V(f)$ (in X_i) all have dimension $d = \dim X - 1 \geq n - 1$. Repeating this process, we arrive at a chain of closed irreducible sets starting in a which has length n . Thus, we have the reverse inequality as well. □

- 2.36 (a) Every Noetherian topological space is compact. In particular, every open subset of an affine variety is compact in the Zariski topology. (Recall that by definition a topological space X is compact if every open cover of X has a finite subcover.)
- (b) A complex affine variety of dimension at least 1 is never compact in the classical topology.

Proof. (a) Let $\{U_i : i \in I\}$ be cover of X , we construct a finite sub cover. Choose any $V_1 \in \{U_i : i \in I\}$ from this cover, if $X = V_1$ we are done. Otherwise, there exists a $V_2 \in \{U_i : i \in I\}$ such that $V_1 \subsetneq V_1 \cup V_2$. Proceeding inductively, we arrive at a increasing chain of open sets:

$$W_n := \bigcup_{i=1}^n U_i.$$

The complements of these sets form a decreasing set of closed sets, and thus since X is Noetherian this chain terminates. This in turn implies that the chain of open sets terminates, which is only possible if after a finite number of iterations, our V_i cover all of X .

- (b) Let X be a complex variety of (Zariski) dimension $r \geq 1$. By Noether's normalization, the ring of polynomial functions on X has the form

$$A(X) = \mathbb{C}[x_1, \dots, x_r][t_1, \dots, t_k]$$

where the x_i are transcendental over $\mathbb{C}$ and the extension

$$\mathbb{C}[x_1, \dots, x_r, t_1, \dots, t_k] / \mathbb{C}[x_1, \dots, x_r]$$

is integral. This yields a surjective map from X to $\mathbb{C}$ which is continuous w.r.t both the Zariski and standard topology, given by

$$x_1 : X \rightarrow \mathbb{C}, p \mapsto x_1(p)$$

Note that it is surjective since $x - \lambda$ is a non constant function on X for all $\lambda \in \mathbb{C}$ and thus has a zero. Since continuous images of compact spaces are compact, X being compact would then yield $\mathbb{C}$ being compact which is not the case.

□

2.40 Let $R = K[x_1, x_2, x_3, x_4] / \langle x_1x_4 - x_2x_3 \rangle$. Show:

- (a) R is an integral domain of dimension 3.
- (b) $x_1, \dots, x_4$ are irreducible, but not prime in R . In particular, R is not a unique factorization domain.
- (c) x_1x_4 and x_2x_3 are two decompositions of the same element of R into irreducibles that do not agree up to units.
- (d) $\langle x_1, x_2 \rangle$ is a prime ideal of codimension 1 in R that is not principal.

In particular, the plane $V(x_1, x_2)$ is a hypersurface in the affine variety $X = V(x_1x_4 - x_2x_3)$ whose defining ideal cannot be generated by one element in $A(X)$.

Proof. (a) To see that R is an integral domain, we will show that the element $f := x_1x_4 - x_2x_3$ is irreducible and hence prime. To do this, we may check that it is irreducible when viewed as a polynomial in the variable x_4 over the ring $K[x_1, x_2, x_3]$, i.e. as an element of $K[x_1, x_2, x_3][x_4]$. When we view f in this way, f is a linear polynomial with trivial content and thus irreducible (recall that the content of a polynomial in one variable is the g.c.d of its coefficients). Thus, we have shown that R is an integral domain. The dimension of R follows immediately from Krull's principal ideal theorem.

- (b) We will show that x_1 is irreducible, the other variables follow analogously. Assume x_1 were reducible, then we would obtain an equation of the form:

$$x_1 = f \cdot g + h \cdot (x_1x_4 - x_2x_3).$$

Now if we consider only the terms which have degree ≤ 1 , and equate, we get

$$x_1 = f_1 g_0 + f_0 g_1 = (f_1 + f_0)(g_1 + g_0)$$

where f_i (resp. g_i) are the homogeneous part of f (resp. g) which have degree i . since f is irreducible, either $f_1 + f_0 = x_1$ or $g_1 + g_0 = x_1$, (up to association). w.l.o.g let $f_1 + f_0 = x_1$ i.e $f_0 = g_1 = 0$ and $g_0 = 1$. Then considering the original equation. The higher order terms arising from $f \cdot g$ must cancel precisely with $h \cdot (x_1 x_4 - x_2 x_3)$ and thus, we may assume in the above equality, that $h = 0$, by leaving out the higher terms of f and g respectively. But then f is associated to f in $K[x_1, \dots, x_4]$ and therefore surely also in R . To observe, that the elements are not prime, we once again only show that x_1 is not prime. In R , we have an equation:

$$x_1 x_2 = x_3 x_4$$

and so x_1 divides the product $x_3 x_4$. But it does not divide x_3 nor x_4 by a similar degree argument as before. Thus we have exhibited elements which are irreducible but not prime.

- (c) That the elements agree is immediate by the definition of R . That these decompositions are different follows from the discussion in (b)
- (d) By the isomorphism theorem, we have

$$R/\langle x_1, x_2 \rangle \simeq \frac{K[x_1, x_2, x_3, x_4]}{\langle x_1, x_2, x_1 x_4 - x_2 x_3 \rangle} = \frac{K[x_1, x_2, x_3, x_4]}{\langle x_1, x_2 \rangle} \simeq K[x_3, x_4]$$

where the last ring is known to have dimension 2. In particular $\langle x_1, x_2 \rangle$ has codimension 1 in R . This ideal is not prime (again in R) since any generator would have to divide both x_1 and x_3 which are known to be irreducible and non-associated by (b).

□

3 Chapter 3: The Sheaf of Regular Functions

3.12 Show that Example 3.11 can be generalized as follows: Let Y be a non-empty irreducible subvariety of an affine variety X , and set $U = X \setminus Y$.

- (a) Assume that $A(X)$ is a unique factorization domain. Show that

$$\mathcal{O}_X(U) = A(X) \iff \text{codim } Y \geq 2.$$

- (b) Show by example that the equivalence of (a) is in general false if $A(X)$ is not assumed to be a unique factorization domain.

Proof. (a) " $\Rightarrow$ ": Let us assume $\text{codim } Y < 2$, we will show that $\mathcal{O}_X(U) \neq A(X)$. If $\text{codim } Y = 0$, then Y corresponds to an irreducible component of X . Since $A(X)$ is a U.F.D and hence an integral domain, X is itself irreducible and therefore $Y = X$ and $U = \emptyset$. The set $\mathcal{O}_X(\emptyset)$ contains only the empty function and hence does not equal $A(X)$, which contains, for example, all of k . If we now assume $\text{codim } Y = 1$, then Y is a hypersurface in X , and since $A(X)$ is a UFD, it is represented by one equation $Y = V(f)$. Due to the fact that we assumed Y to be nonempty, f has a zero in $A(X)$ and is not a unit in $A(X)$. Using the lemma on regular functions on distinguished open sets, we obtain:

$$\mathcal{O}_X(U) = \mathcal{O}_X(D(f)) \simeq A(X)_f \not\simeq A(X).$$

" $\Leftarrow$ ": Let us assume $\text{codim } Y \geq 2$. In this case there exists a strict chain of closed irreducible sets in X of the form

$$Y \subseteq Y_0 \subsetneq Y_1 \subsetneq Y_2 \subseteq X$$

We may assume w.l.o.g that $Y_0 = Y$, that $Y_2 = X$ and that Y_1 is a irreducible hypersurface in X . Since $A(X)$ is a U.F.D, Y_1 may be written as $V(f)$ for prime $f \in A(X)$. Then the above chain may be rewritten as

$$Y \subsetneq V(f) \subsetneq X.$$

Passing to ideals, we have:

$$I(Y) \subsetneq (f) \subsetneq (0).$$

□

since the first inclusion is strict, $I(Y)$ contains a polynomial g which is not divisible by f . Since $I(Y)$ is radical, we may also assume it be squarefree and we have: $I(Y) \supseteq \langle f, g \rangle \subsetneq \langle f \rangle \subsetneq (0)$. This, in turn yields again:

$$Y \subseteq V(f) \cap V(g) \subsetneq V(f) \subsetneq X.$$

Finally, passing to complements:

$$U \supseteq D(f) \cup D(g) \subsetneq D(f) \subsetneq X.$$

Note, that we have a map induced by restrictions of the form

$$\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(D(f) \cup D(g)), \varphi \mapsto \varphi|_{D(f) \cup D(g)}$$

We claim that this map is injective. Indeed, since X is irreducible U is irreducible as well and hence every open subset of U is dense in U . If φ is a regular function on U which restricts to zero, on $D(f) \cup D(g)$, then its zero locus is a closed set containing a nonempty open i.e. it is all of U . If we can show that the equality $\mathcal{O}_X(D(f) \cup D(g)) = A(X)$ holds,

the result would follow for $\mathcal{O}_X(U)$ as well. Suppose now we are given a regular function $\varphi \in \mathcal{O}_X(D(f) \cup D(g))$. Then, φ is regular on both $D(f)$ and $D(g)$, and thus, by the lemma of regular functions on distinguished open sets we have:

$$\begin{aligned}\varphi &\equiv \frac{A}{f^n} \text{ on } D(f) \\ \varphi &\equiv \frac{B}{g^m} \text{ on } D(g)\end{aligned}$$

For some $A, B \in A(X)$. Since $A(X)$ is a UFD, we may assume that f does not divide A and g does not divide B . On the intersection $D(f) \cap D(g)$, these two representations of φ must agree and thus we have an equality:

$$Ag^m = Bf^n \text{ on } D(f) \cap D(g).$$

Now the set in X on which this equality holds is closed and it contains a nonempty open, thus by the irreducibility of X , this equality holds on all of X . In other words, these elements are the same in $A(X)$. Since we have assumed f to be prime and to not divide g , f^n must divide A which is only possible for $n = 0$ by the choice of A . But this implies that

$$\varphi \equiv A \text{ on } D(f).$$

the set in U where this equality holds is closed and contains a nonempty open. Since U is irreducible, the above equality holds in all of U .

- (b) Let us consider again the example from 2.40: $X = V(x_1x_4 - x_2x_3) \subseteq \mathbb{A}^4$. From 2.40, we know that X is an irreducible affine variety whose coordinate ring $\mathcal{O}_X(X)$ is not an integral domain. Furthermore, we have seen that $Y' = V(x_1, x_2) \subseteq X$ is a irreducible subvariety in X of codimension 1. Thus $Y = V(x_1, x_2, x_3)$ is a subvariety of X which has codimension ≥ 2 . We construct a regular function φ on the open set $U := X \setminus Y$ which is not global. Define this function as

$$\varphi : U \rightarrow k, (x_1, x_2, x_3, x_4) \mapsto \begin{cases} \frac{x_4}{x_1} & \text{when } x_1 \neq 0 \\ \frac{x_3}{x_2} & \text{when } x_2 \neq 0 \end{cases}$$

By the definition of X , the two cases agree on their intersections and since φ is locally given by rational functions, it is regular as well. We claim that φ is not the restriction of a global function on X . Indeed, assume we had an equality

$$G = \frac{x_4}{x_1} \text{ when } x_1 \neq 0$$

This would yield $Gx_1 = x_4$ on $U \cap \{x_1 \neq 0\}$. However, the set where this equality holds in X is closed and contains a non-empty open. Thus, this equality holds globally in X . Thus we have a relation $Gx_1 = x_4$ in the ring

$A(X)$. However we have seen in 2.40 that x_1 and x_4 are non-associated irreducible in $A(X)$, thus such an equality is impossible. In particular Y is an example of a codimension ≥ 2 (actually $\text{codim } Y = 2$ by Krull's PID, but we do not need this fact here) irreducible variety in X such that $\mathcal{O}_X(X \setminus Y) \neq A(X)$.

3.20. Let $X \subset \mathbb{A}^n$ be an affine variety, and let $a \in X$. Show that

$$\mathcal{O}_{X,a} \cong \mathcal{O}_{\mathbb{A}^n,a} / I(X) \mathcal{O}_{\mathbb{A}^n,a},$$

where $I(X) \mathcal{O}_{\mathbb{A}^n,a}$ denotes the ideal in $\mathcal{O}_{\mathbb{A}^n,a}$ generated by all quotients $\frac{f}{1}$ with $f \in I(X)$.

Proof. We follow a series of (natural) isomorphisms:

$$\mathcal{O}_{X,a} \cong A(X)_{I(a)} \cong \left(\frac{A(\mathbb{A}^n)}{I(X)} \right)_{I(a)} \cong \frac{(A(\mathbb{A}^n)_{I(a)^c})}{I(X)^e} \cong \mathcal{O}_{\mathbb{A}^n,a} / I(X) \mathcal{O}_{\mathbb{A}^n,a}.$$

The only part of this chain of isomorphisms which was not proven in the notes is the second last isomorphism. This isomorphism follows from the following well known algebraic fact about localizations:

$$S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N).$$

□

3.21. Let a be any point on the real line $\mathbb{R}$. For which of the following sheaves $\mathcal{F}$ on $\mathbb{R}$ (with the standard topology) is the stalk $\mathcal{F}_a$ a local ring (i.e. it has exactly one maximal ideal)?

- (a) $\mathcal{F}$ is the sheaf of continuous functions;
- (b) $\mathcal{F}$ is the sheaf of locally polynomial functions.

Proof. (a) For the first sheaf, the stalk at a point is always local ring. Indeed, let $a \in \mathbb{R}$ and consider the ideal m of functions which vanish at a . This ideal is clearly maximal in $\mathcal{F}_a$ since we have

$$\mathcal{F}_a / m \cong \mathbb{R}.$$

We claim that every element not in this ideal is invertible. Indeed, if (U, φ) is the stalk of a continuous function φ , which does not vanish at a , then we may observe that φ also does not vanish on some neighborhood $V \subseteq U$ of a . On this neighborhood, the function $\frac{1}{\varphi}$ is a well defined continuous function. Therefore, φ has a multiplicative inverse on a region around a and thus its germ is invertible as well.

- (b) For the second sheaf, $\mathcal{F}_a$ is not a local ring. We will show this for the case $a = 0$. Once again the ideal m of stalks vanishing at 0 is maximal, however we will show that not every element in $F_a \setminus m$ is invertible. indeed the polynomial $p = x+1$ does not have a polynomial inverse on any open containing 0, since such an inverse would have to agree with $\frac{1}{x+1}$ which has non-vanishing derivatives of every order at 0, which a property that no polynomial has,

□

3.22. Let $\varphi, \psi \in \mathcal{F}(U)$ be two sections of a sheaf $\mathcal{F}$ on an open subset U of a topological space X . Show:

- (a) If the germs agree in all stalks, i.e. $\varphi_a = \psi_a$ for all $a \in U$, then $\varphi = \psi$.
- (b) If $\mathcal{F} = \mathcal{O}_X$ is the sheaf of regular functions on an irreducible affine variety X , then we can already conclude $\varphi = \psi$ if there exists $a \in U$ with $\varphi_a = \psi_a$.
- (c) For a general sheaf $\mathcal{F}$ on a topological space X , the statement in (b) is false.

Proof. (a) Suppose we have $\varphi_a = \psi_a$ for all $a \in U$. Then for all $a \in U$, there is an open set U_a containing a such that $\varphi|_{U_a} = \psi|_{U_a}$. The sets $\{U_a : a \in U\}$ form an open cover of U and the sections $\beta_a \in \mathcal{F}(U_a)$ where $\beta_a := \varphi|_{U_a} = \psi|_{U_a}$ may be glued together to a unique global section on U . The uniqueness of this global section forces, $\varphi = \psi$.

- (b) Suppose now that $\mathcal{F} = \mathcal{O}_X$ and there is a point $a \in U$ such that $\varphi_a = \psi_a$. Then there exists a open subset $V \subseteq U$ such that $\varphi - \psi \equiv 0$ on V . We have assumed X to be irreducible, and thus U is irreducible as well. Furthermore, we know that the set where $\varphi - \psi \equiv 0$ holds is closed and therefore, since it contains nonempty open, it is all of U .
- (c) Let $X = \mathbb{A}_{\mathbb{C}}^1$ and consider the sheaf of $\mathbb{C}$ -valued functions on X (not necessarily continuous). The the two global sections given by

$$\begin{aligned}\varphi(x) &= x \\ \psi(x) &= \begin{cases} x & \text{if } \mathcal{I}(x) > 0 \\ -x & \text{if } \mathcal{I}(x) \leq 0 \end{cases}\end{aligned}$$

have matching stalks at all x on the upper half plane, however globally they disagree.

□

3.23. Let $\mathcal{F}$ be a sheaf on a topological space X , and let Y be a non-empty irreducible closed subset of X . Define the stalk of $\mathcal{F}$ at Y by

$$\mathcal{F}_Y := \{(U, \varphi) : U \subset X \text{ open, } U \cap Y \neq \emptyset, \varphi \in \mathcal{F}(U)\} / \sim,$$

where $(U, \varphi) \sim (U', \varphi')$ iff there exists an open set $V \subset U \cap U'$ with $V \cap Y \neq \emptyset$ and $\varphi|_V = \varphi'|_V$. It therefore describes functions in an arbitrarily small neighborhood of an arbitrary dense open subset of Y . If Y is a non-empty irreducible subvariety of an affine variety X , prove that the stalk $\mathcal{O}_{X,Y}$ of $\mathcal{O}_X$ at Y is a K -algebra isomorphic to the localization $A(X)_{(Y)}$ (hence giving a geometric meaning to this algebraic localization).

Proof. Let us exhibit the (natural) isomorphism. Recall first, that $A(X)_{(Y)}$ is the set of formal fractions $\frac{f}{g}$ where $g \notin I(Y)$. For any such fraction, the Nullstellensatz implies that $D(g) \subseteq X$ must intersect Y . Since Y is irreducible, this open subset is dense in Y . This allows us to define the map Φ as

$$\Phi : A(X)_{(Y)} \rightarrow \mathcal{O}_{X,Y}, \quad \frac{f}{g} \mapsto [D(g), \frac{f}{g}].$$

First, let us prove that it is well defined. suppose we have an equality of formal fractions $\frac{A}{B} = \frac{C}{D}$ in $A(X)_{(Y)}$. Then, there exists an element $s \in A(X) \setminus I(Y)$, such that:

$$s(AD - BC) = 0.$$

Since $s \notin I(Y)$, the open $D(s)$ must intersect Y and is therefore also dense. Since the three open sets $D(s), D(A), D(B)$ are all dense, and Y is irreducible, their intersection $D(s) \cap D(A) \cap D(B)$ is also a open subset which is dense in Y . On this set, we have $s(AD - BC) = 0$ but $s \neq 0$, and so $AD - BC = 0$ which implies that on this open subset $\frac{A}{B} = \frac{C}{D}$. In particular, we have proven

$$(D(B), \frac{A}{B}) \sim (D(D), \frac{C}{D})$$

which proves that Φ is well-defined. From, this we also see that Φ is a morphism of K -algebra. Next, we prove that Φ is injective. Suppose we have

$$(D(g), \frac{f}{g}) \sim (Y, 0).$$

Then there exists a open subset V , intersecting Y non trivially, such $\frac{f}{g} \equiv 0$ on V . Since this is a closed relation, we find that this actually holds on the irreducible component X' of X which contains Y . On the, other components, $X_1, \dots, X_r$ of X , choose elements $s_1, \dots, s_r \in A(X)$ which vanish on X_r but do not vanish on Y (which is possible due to the Nullstellensatz again). Multiplying these terms together we see that $f \cdot s_1 \cdots s_r$ vanishes on every component of X and thus on all of X . In particular:

$$s_1 \cdots s_r (f \cdot 1 - g \cdot 0) \equiv 0,$$

which is exactly the condition for $\frac{f}{g} = \frac{0}{1}$ to hold in $A(X)_{(Y)}$. □

- 3.24. Let $\mathcal{F}$ be a sheaf on a topological space X , and let $a \in X$. Show that the stalk $\mathcal{F}_a$ is a local object in the following sense: if $U \subset X$ is an open neighborhood of a , then $\mathcal{F}_a$ is isomorphic to the stalk of $\mathcal{F}|_U$ at a on the topological space U .

Proof. Consider the map

$$\Phi : \mathcal{F}_a \rightarrow (\mathcal{F}|_U)_a, [(V, \varphi)] \mapsto [(U \cap V, \varphi|_{U \cap V})]$$

This map is clearly well defined, since if two sections agree on a neighborhood containing a , then this neighborhood may be shrunk to be contained in U , such that they agree on a neighborhood containing a which is contained in U . We will prove that Φ is an isomorphism, when $\mathcal{F}$ takes values in the category of sets. It is clear that if $\mathcal{F}$ takes values in a category with more structure such as rings, k -Algebras or R -modules, then the above Φ is a morphism in this category. Thus, the general case follows from this special case. To see that Φ is an isomorphism, one may verify that the map

$$\Psi : (\mathcal{F}|_U)_a \rightarrow \mathcal{F}_a, [(V, \varphi)] \mapsto [(V, \varphi)] \in \mathcal{F}_a$$

is a well defined inverse. $\square$

4 Chapter 4: Morphisms

- 4.12. (**Affine conics**) An irreducible quadric curve in $\mathbb{A}^2$ is also called an *affine conic*. Show that every affine conic over a field K with $\text{char } K \neq 2$ is isomorphic to exactly one of

$$X_1 = V(x_2 - x_1^2) \quad \text{or} \quad X_2 = V(x_1 x_2 - 1),$$

with an isomorphism given by a linear coordinate transformation followed by a translation.

Proof. Let us assume that our conic C is given by an equation:

$$C = V(ax^2 + by^2 + cxy + dx + ey + f)$$

Since C is a conic, not all of the elements a, b, c can be zero. We claim that we may assume $a \neq 0$. Indeed, if $b \neq 0$ then the affine transformation $(x, y) \mapsto (y, x)$ makes the coefficient of x^2 nonzero. Similarly, if $a = b = 0$ but $c \neq 0$, then the linear change $(x, y) \mapsto (x, y + x)$ makes the coefficient of x^2 nonzero. This, in turn, implies that we may change coordinates via $(x, y) \mapsto (x/\sqrt{a}, y)$ to assume that the coefficient of x^2 is actually 1. We have thus successfully, transformed our conic to a form:

$$C = V(x^2 + ay^2 + bxy + cx + dy + e). \quad (1)$$

At this point two cases can occur: $a = 0$ or $a \neq 0$, let us investigate the latter first. In this case, we may once again change coordinates to assume $a = 1$ and so our conic is of the form:

$$C = V(x^2 + y^2 + axy + bx + cy + d) = V\left(\left(x + \frac{b}{2}\right)^2 + \left(y + \frac{c}{2}\right)^2 + axy + \tilde{d}\right)$$

Via the obvious translation, this is transformed to

$$C = V(x^2 + y^2 + axy + d) = V((x + ay)^2 - \alpha y^2 + d)$$

Via the substitution $x \rightarrow x + ay$ and $y \rightarrow y$, this is once again transformed to a conic of the form

$$C = V(x^2 + \alpha y^2 + d)$$

we know that $a \neq 0$ since otherwise C would be reducible. Thus, rescaling y we arrive at a form $V(x^2 + y^2 + d)$. now d must be nonzero, since otherwise the above equation factors $x^2 + y^2 = (x + iy)(x - iy)$ where i is a square root of -1 in k . Thus rescaling both x and y by $\frac{i}{\sqrt{d}}$, we arrive at

$$C = V(x^2 + y^2 - 1) = V((x + iy)(x - iy) - 1).$$

Finally, shifting coordinates $x \rightarrow x + iy$, and $y \rightarrow x - iy$, we arrive at the form

$$C = V(xy - 1).$$

To finish the proof, we must revisit in equation (1) the case where $a = 0$ i.e our conic is of the form:

$$C = V(x^2 + bxy + cx + dy + e) = V\left(\left(x + \frac{c}{2}\right)^2 + bxy + dy + e\right).$$

Via a translation in the x -coordinate, this can be transformed to

$$C = V(x^2 + axy + by + d) = V\left(\left(x + \frac{a}{2}y\right)^2 + \gamma y^2 + by + d\right).$$

once again the change of variables $x \rightarrow x + \frac{a}{2}y$ turns this to

$$C = V(x^2 + ay^2 + by + d).$$

Now two cases here can occur, $a = 0$ or $a \neq 0$. In the latter, we arrive to the first case discussed before. If $a = 0$, then $b \neq 0$ and thus the rescaling y by $\frac{1}{b}$, we arrive at a conic of the form

$$C = V(x^2 + (y + d)).$$

Finally, changing the y variable by $y \rightarrow -(y + d)$, we arrive at $C = V(x^2 - y)$ as desired. $\square$

Remark. We crucially used that k was algebraically closed and that $\text{char } K \neq 2$. If we drop either of these assumptions, the claim is false; See for example the classification of conics over $\mathbb{R}$.

4.13. Let $f: X \rightarrow Y$ be a morphism of affine varieties and $f^*: A(Y) \rightarrow A(X)$ the corresponding homomorphism of coordinate rings. Are the following statements true or false?

- (a) f is surjective iff f^* is injective.
- (b) f is injective iff f^* is surjective.
- (c) If $f: \mathbb{A}^1 \rightarrow \mathbb{A}^1$ is an isomorphism then f is affine linear, i.e. $f(x) = ax + b$ for some $a, b \in K$.
- (d) If $f: \mathbb{A}^2 \rightarrow \mathbb{A}^2$ is an isomorphism then f is affine linear, i.e. $f(x) = Ax + b$ for some $A \in \text{Mat}(2 \times 2, K)$ and $b \in K^2$.

Proof. (a) The statement is false in general. Let $X = V(xy - 1) \subseteq \mathbb{A}^2$ and consider the map

$$f: X \rightarrow \mathbb{A}^1, (x, y) \mapsto x.$$

This map is a morphism since its components are given by regular functions on X . Furthermore, f is not surjective since its image does not meet $0 \in \mathbb{A}^1$. The pullback morphism f^* of f is injective. Indeed, suppose $f^*p \equiv 0 \in A(X)$. Then p must vanish on the image of f , since for a point $P \in X$, we have $(f^*p)(P) = p(f(P))$. It is easy to verify, that the image of f is $\mathbb{A}^1 \setminus \{0\}$, and thus p must vanish on the closure of this set which is all of $\mathbb{A}^1$. Thus, we have shown

$$f^*p = 0 \Rightarrow p \equiv 0.$$

- (b) The Statement is false in general. Consider the affine variety $X = V(x^2 - y^3) \subseteq \mathbb{A}^2$. As we have seen in the notes, the morphism

$$f: \mathbb{A}^1 \rightarrow X, p \mapsto (p^3, p^2)$$

is a bijection. However, its pullback morphism is given by

$$f^*: \frac{k[x, y]}{(x^2 - y^3)} \rightarrow k[t], \begin{cases} x \mapsto t^3 \\ y \mapsto t^2 \end{cases}$$

which is not surjective, since t is not in the image.

- (c) The statement is true.

The isomorphisms of $\mathbb{A}^1 \rightarrow \mathbb{A}^1$ are in one to one correspondence to k -algebra isomorphisms $k[t] \rightarrow k[t]$ via the pullback functor. Therefore, we investigate the k -algebra isomorphism of $k[t]$. Any such isomorphism f is uniquely defined by its image of t , let us denote this polynomial as p . If p were a constant, the image of f would be entirely contained in k , and thus f would not be an isomorphism. Similarly if the degree of f would be larger than 1, the polynomials in the image of f would only have nonzero terms at those powers of t which are a multiple of $d := \deg p$. In particular, t itself would not

be in the image of f , and f would not be bijective. Thus p is of the form $At + b$ for a nonzero A . This k -algebra morphism is induced by the affine linear

$$\varphi: \mathbb{A}^1 \rightarrow \mathbb{A}^1, p \mapsto Ap + b$$

- (d) The claim is false. A simple counterexample can be constructed using the k -algebra isomorphism f given by

$$f: k[x, y] \rightarrow k[x, y], \begin{cases} x \mapsto x + y^2 \\ y \mapsto y. \end{cases}$$

This map a morphism as its components are given by regular functions. Furthermore, it has an inverse:

$$g: k[x, y] \rightarrow k[x, y], \begin{cases} x \mapsto x - y^2 \\ y \mapsto y. \end{cases}$$

The k -algebra morphism f , is induced by the isomorphism:

$$F: \mathbb{A}^2 \rightarrow \mathbb{A}^2, (x, y) \mapsto (x + y^2, y)$$

□

4.19. Which of the following ringed spaces are isomorphic over $\mathbb{C}$?

- (a) $\mathbb{A}^1 \setminus \{1\}$
- (b) $V(x_1^2 + x_2^2) \subset \mathbb{A}^2$
- (c) $V(x_2 - x_1^2, x_3 - x_1^3) \setminus \{0\} \subset \mathbb{A}^3$
- (d) $V(x_1 x_2) \subset \mathbb{A}^2$
- (e) $V(x_2 - x_1^3 - x_1^2) \subset \mathbb{A}^2$
- (f) $V(x_1^2 - x_2^2 - 1) \subset \mathbb{A}^2$

We can reduce the task to comparing coordinate rings, by showing that all the ringed spaces shown above are actually affine. This is obvious for all of the spaces listed, except for (a) and (c). For (a) we may write $\mathbb{A}^1 \setminus \{1\} = D(f) \subseteq \mathbb{A}^1$ where $f = x - 1$ and for (c) we may write

$$V(x_2 - x_1^2, x_3 - x_1^3) \setminus \{0\} = D(g) \subseteq V(x_2 - x_1^2, x_3 - x_1^3)$$

where $g = x_1 \in A(V(x_2 - x_1^2, x_3 - x_1^3))$.

Now, that we have shown that all the spaces are affine, we are ready to complete the task. First, we show that (a) is isomorphic to (c). For this, we first show that $V(x_2^2 - x_1^2, x_3 - x_1^3) \cong \mathbb{A}^1$. Indeed, consider the map:

$$f: \mathbb{A}^1 \rightarrow V(x_2^2 - x_1^2, x_3 - x_1^3), x \mapsto (x, x^2, x^3).$$

It is a morphism as its components are regular. It is an isomorphism since projection to the first coordinate defines an inverse. Under this isomorphism, the set $V(x_2 - x_1^2, x_3 - x_1^3) \setminus \{0\}$ maps to $\mathbb{A}^1 \setminus \{0\}$ which is isomorphic to $\mathbb{A}^1 \setminus \{1\}$ via the translation $x \mapsto x + 1$.

Next, we claim that (b) and (d) are isomorphic to one another. This becomes evident once we factor:

$$V(x_1^2 + x_2^2) = V((x_1 + ix_2)(x_1 - ix_2)).$$

we obtain the desired isomorphism Φ as

$$\Phi: V(xy) \rightarrow V((x_1 + ix_2)(x_1 - ix_2)), (x, y) \mapsto \left(\frac{x+y}{2}, \frac{x-y}{2i}\right).$$

Note that these varieties are not isomorphic to (a) since they are reducible and (a) is irreducible as a open subset of a irreducible space. Let us consider next (e). (e) is isomorphic to $\mathbb{A}^1$ via the map

$$\Phi: \mathbb{A}^1 \rightarrow V(x_2 - x_1^3 - x_1^2), x \mapsto (x, x_1^3 + x_1^2)$$

whose inverse map is given by projection. In particular, it is not isomorphic to (a) nor (c) (compare for example units in the coordinate rings). (e) is also irreducible, it is not isomorphic to the line pairs in (b) and (d). Finally, consider the last affine variety in (f). First, we may notice that this affine variety is actually irreducible, since its defining equation is irreducible ($x_2^2 - 1$ is not a square in $k[x_2]$). Thus, (f) corresponds to one of the two conics from 4.12. We claim that it does not correspond to $V(x_1 - x_2^2)$. If it did, it would have the same coordinate ring. One may easily notice that $V(x_1 - x_2^2) \cong \mathbb{A}^1$ (via projection), and thus, in the case of an isomorphism, we would have $A(V(x_1^2 - x_2^2 - 1)) \cong k[t]$. This is not the case however; The ring $R = \frac{k[x,y]}{(x_1^2 - x_2^2 - 1)}$ has nontrivial units (meaning not in k):

$$(x_1 + x_2)(x_1 - x_2) \equiv 1,$$

where neither factor is in k . On the other hand $k[t]$ has no nontrivial units. Thus $V(x_1^2 - x_2^2 - 1) \not\cong V(x_1x_2 - 1)$ where the last set is known to be isomorphic to (a) via projection.

5 Chapter 5: Varieties

5.7. Show:

- (a) Every morphism $f: \mathbb{A}^1 \setminus \{0\} \rightarrow \mathbb{P}^1$ can be extended to a morphism $\mathbb{A}^1 \rightarrow \mathbb{P}^1$.
- (b) Not every morphism $f: \mathbb{A}^2 \setminus \{0\} \rightarrow \mathbb{P}^1$ can be extended to a morphism $\mathbb{A}^2 \rightarrow \mathbb{P}^1$.
- (c) Every morphism $f: \mathbb{P}^1 \rightarrow \mathbb{A}^1$ is constant.

Proof. (a) Recall that we have defined $\mathbb{P}^1$ as the gluing of $X_1 = \mathbb{A}^1 = X_2$ at the open sets $U_{1,2} = \mathbb{A}^1 \setminus \{0\} = U_{2,1}$ along the map

$$g : U_{1,2} \rightarrow U_{2,1}, x \mapsto \frac{1}{x}.$$

Not both of the sets

$$U := f^{-1}(X_1)$$

$$V := f^{-1}(X_2)$$

can be empty since $\mathbb{P}^1 = X_1 \cup X_2$. Let us assume f is nonconstant, then both are nonempty (in the constant case, it is clear how to extend f). Then U is a nonempty open subset of $\mathbb{A}^1 \setminus \{0\}$ and thus it is given by

$$U = \mathbb{A}^1 \setminus \{0, p_1, \dots, p_k\}.$$

Since f restricted to this open set is a morphism to the affine variety $X \cong \mathbb{A}^1$, it is given by a regular function on this set. This is a distinguished open set, namely it is the zero locus of the polynomial

$$P = x \cdot \prod_{i=1}^k (x - p_k),$$

and therefore we know its ring of regular functions: $\mathcal{O}_{\mathbb{A}^1}(U) = k[x]_P$. Thus, on the open set U we may write

$$f = \frac{A}{x^n \cdot \left(\prod_{i=1}^k (x - p_k)\right)^k}$$

where $A \in k[x]$ such that x does not divide A . Now if $n = 0$, we can notice that this expression is defined at zero as well. Infact, this is then a regular function on $U \cup \{0\}$. Thus, extending f in this way to zero, yields the extension in mind. Notice that it is a morphism since it is a morphism when restricted either of the open sets $U \cup \{0\}$ and V . Now if $n > 0$, then let us define $f(0) = \infty$. Once again we may verify that this makes f a morphism since it is a morphism when restricted to U and when restricted to $V \cup \{0\}$.

- (b) We exhibit a morphism $f : \mathbb{A}^2 \setminus \{(0,0)\} \rightarrow \mathbb{P}^1$ which can't be extended to zero. Once again, we represent $\mathbb{P}^1$ as the gluing of two affine sets $X_1 = \mathbb{A}^1 = X_2$, along the map

$$g : \mathbb{A}^1 \setminus \{0\} \rightarrow \mathbb{A}^1 \setminus \{0\}, x \mapsto \frac{1}{x},$$

and we denote the inclusions as $i_j : X_j \rightarrow \mathbb{P}^1$. Consider the two maps

$$\begin{aligned} f_1 : \mathbb{A}^2 \setminus V(x) &\rightarrow X_1 (x, y) \mapsto \frac{y}{x} \\ f_2 : \mathbb{A}^2 \setminus V(y) &\rightarrow X_2 (x, y) \mapsto \frac{x}{y} \end{aligned}$$

As regular functions, these are morphisms. This allows us to define the two morphisms

$$\begin{aligned} i_1 \circ f_1 : \mathbb{A}^2 \setminus V(x) &\rightarrow \mathbb{P}^1 (x, y) \mapsto i_1 \left(\frac{y}{x} \right) \\ i_2 \circ f_2 : \mathbb{A}^2 \setminus V(y) &\rightarrow \mathbb{P}^1 (x, y) \mapsto i_2 \left(\frac{x}{y} \right) \end{aligned}$$

We can observe, that these two morphisms agree on the intersection of their domains: $\mathbb{A}^2 \setminus V(xy)$ and their domains cover $\mathbb{A}^2 \setminus \{(0, 0)\}$. Thus, they define a well defined morphism $f : \mathbb{A}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{P}^1$. It remains to show, that f cannot be extended to zero. We may note that f maps $V(x) \setminus \{(0, 0)\}$ entirely to $0 \in \mathbb{P}^1$. Thus, if we could extend f to zero, f would have to map $(0, 0)$ to 0 as well since otherwise the image of $V(x)$ would not be irreducible. However, making the same considerations for $V(y)$, we see that the image of $(0, 0)$ must be ∞ , and thus there is no value we can choose for $f(0, 0)$ which would extend f as a morphism.

- (c) We take the same representation of $\mathbb{P}^1$ as in part (a) and (b). Suppose we are given a morphism $f : \mathbb{P}^1 \rightarrow \mathbb{A}^1$. Then, this morphism restricts to two morphisms:

$$\begin{aligned} f_1 : X_1 &\rightarrow \mathbb{A}^1 \\ f_2 : X_2 &\rightarrow \mathbb{A}^1. \end{aligned}$$

Since $X_1 \cong \mathbb{A}^1$, f_1 is given by a polynomial $p(x) \in k[x]$ where x is a coordinate of $X_1 \cong \mathbb{A}^1$. Similarly, f_2 is given by a polynomial $g(t) \in k[t]$ where t is a coordinate of $X_2 \cong \mathbb{A}^1$. If we evaluate f_2 on a point $t \in X_2 \setminus \{0\}$, we obtain:

$$f_2(t) = f(i_2(t)) = f\left(i_1\left(\frac{1}{t}\right)\right) = f_1\left(\frac{1}{t}\right) = p\left(\frac{1}{t}\right).$$

on the otherhand, this expression should match with $g(t)$:

$$p\left(\frac{1}{t}\right) = g(t) \text{ on } \mathbb{A}^1 \setminus \{0\}.$$

if $\deg(p) = d$, then we may represent $p\left(\frac{1}{t}\right) = \frac{A}{t^d}$, where t does not divide A . Multiplying the above relation by t^d , we obtain $A = t^d g(t)$. Notice that this is a closed condition, and thus this equality actually holds globally, in $k[t]$, which is only possible for $d = 0$. Therefore, p was constant and therefore f was as well.

□

- 5.8. (a) Show that every isomorphism $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is of the form

$$f(x) = \frac{ax + b}{cx + d} \quad \text{for some } a, b, c, d \in K,$$

where x is an affine coordinate on $\mathbb{A}^1 \subset \mathbb{P}^1$.

- (b) Given three distinct points $a_1, a_2, a_3 \in \mathbb{P}^1$ and three distinct points $b_1, b_2, b_3 \in \mathbb{P}^1$, show that there is a unique isomorphism $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ with $f(a_i) = b_i$ for $i = 1, 2, 3$.

Proof. (a) We distinguish into two cases; $f(\infty) = \infty$ and $f(\infty) \neq \infty$. In the first case, f restricts to an isomorphism

$$f|_{X_1} : \mathbb{A}^1 \cong X_1 \rightarrow X_1.$$

Thus, from 4.13 (c), f can be represented on this subset as $f = ax + b$ as desired. Now let assume we have $a, b \in \mathbb{A}^1$ with $f(a) = \infty$ and $f(\infty) = b$. In this case, f restricts to an isomorphism of the form:

$$f|_{\mathbb{A}^1 \setminus \{a\}} : \mathbb{A}^1 \setminus \{a\} \rightarrow \mathbb{A}^1 \setminus \{b\} \subseteq \mathbb{A}^1.$$

These are both affine varieties and so f is given by a singular regular function on $\mathbb{A}^1 \setminus \{a\}$. In particular we have $f = \frac{A}{(x-a)^n}$ where $A \in k[x]$. We may furthermore assume, that $x - a$ does not divide A . If for the reduced A we have $\deg(A) > 1$, then f is not injective since it has multiple zeros (not equal to a). Similarly, $n = 0$, since $1/f$ is an isomorphism outside of the roots of A . Thus f is of the form

$$f = \frac{\lambda x - \beta}{x - a} \text{ for some } \lambda \neq 0, \beta \neq a.$$

Which is equivalent to the form we wanted to achieve.

- (b) We will first show, that there is a unique mapping which maps the points to a_1, a_2, a_3 to $0, 1, \infty$ respectively. In this case there is also a unique inverse to this map. The desired unique map would be obtained by mapping a_1, a_2 and a_3 to $0, 1, \infty$ and then mapping these points to b_1, b_2, b_3 . For existence, let us assume a_1, a_2, a_3 are all affine (if not change coordinates to a set where this is true and find that in these coordinates f exists and is unique).

$$f(x) = \frac{(a_2 - a_3)(x - a_1)}{(a_2 - a_1)(x - a_3)}$$

Now, let us address uniqueness. Any function which satisfies these properties is represented by

$$f = \frac{ax + b}{cx + d}$$

of course we may scale this such that $c = 1$ without changing the function (notice $c \neq 0$ since $f(a_3) = \infty$. so we have such that we have

$$\begin{cases} \frac{aa_1+b}{a_1+d} = 0 \\ \frac{aa_2+b}{a_2+d} = 1 \\ a_3 - d = 0 \end{cases}.$$

Multiplying out the denominators, we arrive at a system of three linear equations in the variables a, b, d , and thus there is at most one solution.

□

- 5.9. If X and Y are affine varieties, there is a bijection (Proposition 3.8 and Corollary 4.8)

$$\begin{aligned} \{\text{morphisms } X \rightarrow Y\} &\xleftrightarrow{\text{11}} \{K\text{-algebra homomorphisms } \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)\}, \\ f &\longmapsto f^*. \end{aligned}$$

Does this statement still hold

- (a) if X is an arbitrary prevariety (but Y is still affine);
- (b) if Y is an arbitrary prevariety (but X is still affine)

Proof. (a) The claim does not hold in general. To see this, let $X = \mathbb{A}^2 \setminus \{0\}$ and $Y = \mathbb{A}^2$. We have already seen that the inclusion of X into Y induces the identity isomorphism of coordinate rings although it is not an isomorphism of prevarieties. (in particular, any other inclusion also induces the same isomorphism).

- (b) The claim does not hold in general. Let $X = \mathbb{A}^1$ and $Y = \mathbb{P}^1$. As we have seen in the 5.7 (c), $\mathcal{O}_Y(Y) = k$. Thus there is only one morphism from $\mathcal{O}_Y(Y) = k$ to $\mathcal{O}_X(X) = k[t]$, namely inclusion of k . However, as we have seen in 5.8, there are many different morphism from $\mathbb{A}^1$ to $\mathbb{P}^1$ (which given by restricting the different functions f in 5.7 to $\mathbb{A}^1$).

□

- 5.11. Let Y be a closed subset of a prevariety X , considered as a ringed space with the structure sheaf of Construction 5.10 (b). Prove for every affine open subset $U \subset X$ that the ringed space $U \cap Y$ (considered as an open subset of the ringed space Y as in Definition 4.1 (c)) is isomorphic to the affine variety $U \cap Y$ (considered as an affine subvariety of the affine variety U).

In particular, this shows that Construction 5.10 (b) makes Y into a prevariety, and that this prevariety is isomorphic to the affine variety Y if X is itself affine (and thus Y an affine subvariety of X).

Proof. The natural map between these two space is of course the identity, we only need to show that it is a morphism of ringed spaces in both directions. For an open set $V \subseteq Y \cap U$, the pullback of f is given by

$$f^* : \mathcal{O}_{U \cap Y}(V) \rightarrow \mathcal{O}_Y(V), \quad g \mapsto g.$$

Thus, we need to prove that if g is regular on V with respect to the structure sheaf $\mathcal{O}_{U \cap Y}$, then it is also regular on V with respect to the structure sheaf $\mathcal{O}_Y$. This is essentially obvious; The regular functions on V w.r.t $\mathcal{O}_Y$ are given by those functions from V to k which are locally at every point, represented as restrictions of regular functions w.r.t the sheaf $\mathcal{O}_X$. On the otherhand, the functions $\mathcal{O}_U$ carries the trace sheaf of X , its sections agree with those of X . Since U is affine, the sections of $\mathcal{O}_{U \cap Y}$ were defined to be restrictions of those of $\mathcal{O}_U$ and thus sections of X . In particular, locally at every point, they look like regular functions on X on small open subsets. It remains to show that

$$g^* : \mathcal{O}_Y(V) \rightarrow \mathcal{O}_{U \cap Y}(V), g \mapsto g,$$

is also a morphism. We want to show that regular functions w.r.t $\mathcal{O}_Y$ are also regular w.r.t $\mathcal{O}_{U \cap Y}$. The regular functions of $\mathcal{O}_{U \cap Y}(V)$ are given by functions from V to k , which are locally restrictions of functions in $\mathcal{O}_U(V)$ to Y . Using the fact that, $\mathcal{O}_U(V) = \mathcal{O}_X(V)$, we see that these are all locally represented by regular functions w.r.t the structure sheaf of X . Since Y can be covered by affines, it follows that Y is a prevariety w.r.t this construction. $\square$

5.21. Show that the space $\mathbb{P}^1$ of Example 5.5 (a) is a variety.

Proof. Recall that $\mathbb{P}^1$ was defined as the gluing of two affine lines $X_1 \cong \mathbb{A}^1 \cong X_2$ along the open sets $\mathbb{A}^1 \setminus \{0\}$ via the map the gluing map $x \mapsto \frac{1}{x}$. Let us denote by

$$i_1 : X_1 \rightarrow \mathbb{P}^1$$

the respective inclusion maps. The affine open sets $i_i(X_i)$ for $i = 1, 2$ form an open cover of $\mathbb{P}^1$. Indulging in a slight abuse of notation, we will write X_i for $i_i(X_i)$ as well. The product $\mathbb{P}^1 \times \mathbb{P}^1$ was defined as the gluing:

$$(X_1 \times X_1 \cup X_1 \times X_2 \cup X_2 \times X_2 \cup X_2 \times X_1) / \sim$$

Where we identify $(x_1, y_1) \sim (x_2, y_2)$ if and only if $x_1 = x_2$ and $y_1 = y_2$ in $\mathbb{P}^1$. We already know that $\mathbb{P}^1 \times \mathbb{P}^1$ is a prevariety, so we only verify the hausdorff axiom. The above space carries the quotient topology, and thus a subset $C \subseteq X$ is closed if and only if the preimage under the four inclusions:

$$\begin{aligned} i_{1,1} : X_1 \times X_1 &\rightarrow \mathbb{P}^1 \times \mathbb{P}^1 \\ i_{1,2} : X_1 \times X_2 &\rightarrow \mathbb{P}^1 \times \mathbb{P}^1 \\ i_{2,1} : X_2 \times X_1 &\rightarrow \mathbb{P}^1 \times \mathbb{P}^1 \\ i_{2,2} : X_2 \times X_2 &\rightarrow \mathbb{P}^1 \times \mathbb{P}^1 \end{aligned}$$

let us check this for the diagonal $\Delta \subseteq \mathbb{P}^1 \times \mathbb{P}^1$. We have:

$$\begin{aligned} i_{1,1}^{-1}(\Delta) &= X_1 \times X_1 \cong \mathbb{A}^1 \times \mathbb{A}^1 \cong \mathbb{A}^2 \\ i_{1,2}^{-1}(\Delta) &\cong \{(x, y) \in \mathbb{A}^1 \times \mathbb{A}^1 \mid y = \frac{1}{x}\} \cong V(xy - 1), \end{aligned}$$

which are both closed sets in their respective spaces. Similarly, the preimages of the diagonal under $i_{2,2}$ and $i_{2,1}$ are both closed. Since all four preimage are closed, Δ is closed as well. $\square$

5.22. Let X and Y be prevarieties. Show:

- (a) If X and Y are varieties then so is $X \times Y$.
- (b) If X and Y are irreducible then so is $X \times Y$.

Proof. (a) We are asked to verify the Hausdorff axiom for the product of two varieties X and Y . Note that if X and Y are both affine, then so is $X \times Y$ and thus it is a variety as well. For the general case, let us cover X by open affine sets $U_1, \dots, U_n$ and Y is covered by $V_1 \dots V_m$. The affine sets $U_i \times V_j$ cover the space $X \times Y$. This means, that the space

$$(X \times Y) \times (X \times Y)$$

is covered by sets of the form $(U_i \times V_j) \times (U_{i'} \times V_{j'})$ and the diagonal is closed in $X \times Y$ if and only if its intersection with $(U_i \times V_j) \times (U_{i'} \times V_{j'})$ is always closed (in this affine space). For this we have:

$$\Delta \cap (U_i \times V_j) \times (U_{i'} \times V_{j'}) \cong (\Delta_X \cap (U_i \times U_{i'})) \times (\Delta_Y \cap (V_j \times V_{j'}))$$

where we used a change in the order of variables. The above presentation is clearly closed since the Δ_X and Δ_Y .

- (b) Since X and Y are irreducible, every open subset of these spaces are dense. In particular, we cover X by affine open subsets $U_i \subseteq X$ and Y by $V_j \subseteq Y$. By 2.20 they are irreducible and by 2.24, their product $U_i \cap V_j$ is irreducible. By the construction of $X \times Y$, the product $X \times Y$ can be covered by such open sets. Lastly we have:

$$(U_i \times V_j) \cap (U_{i'} \times V_{j'}) = (U_i \cap U_{i'}) \times (V_j \cap V_{j'}) \neq \emptyset,$$

since the irreducibility of X and Y ensures that any two nonempty open sets intersect. Finally, 2.21 proves that $X \times Y$ is irreducible in this case as well. $\square$

5.23. Use diagonals to prove the following statements:

- (a) The intersection of any two affine open subsets of a variety is again an affine open subset.

- (b) If $X, Y \subset \mathbb{A}^n$ are two pure-dimensional affine varieties then every irreducible component of $X \cap Y$ has dimension at least $\dim X + \dim Y - n$.

Proof. (a) We claim that $U \cap V \cong \Delta_X \cap (U \times V)$. For this, note that the map

$$i : X \rightarrow \Delta_X \subseteq X \times X, x \mapsto (x, x)$$

is a morphism by the universal property of the product. This restricts to a morphism of $U \cap V \rightarrow \Delta_X$. The image of this set is precisely, they set $\Delta_X \cap (U \times V)$. This set is a closed subvariety of the affine variety $U \times V$ and thus affine. Furthermore, projection to the first coordinate defines an inverse to i , and thus i is an isomorphism.

- (b) As we have seen in (a), we have the useful isomorphism :

$$i : X \cap Y \rightarrow \Delta_{\mathbb{A}^n} \cap (X \times Y) \subseteq X \times Y \subseteq \mathbb{A}^n \times \mathbb{A}^n$$

We already know that for the affine product, we have $\dim(X \times Y) = \dim X + \dim Y$. Now if $x_1, \dots, x_n$ are coordinates for X and $y_1, \dots, y_n$ for Y , then we have:

$$\Delta_{\mathbb{A}^n} \cap (X \times Y) = V(x_1 - y_1, \dots, x_n - y_n) \subseteq X \times Y.$$

Thus, by Krulls principle ideal theorem, the dimension of $\Delta_{\mathbb{A}^n} \cap (X \times Y)$ is at least $\dim(X \times Y) - n = \dim X + \dim Y - n$

□

- 5.24. In Exercise 2.34 (b) we have seen that the dimension of a dense open subset U of a topological space X need not be the same as that of X . However, show now that $\dim U = \dim X$ holds in this situation if X is a variety.

Proof. We may Assume X to be irreducible without loss of generality. Indeed if U is dense in X , then it is dense in a irreducible component of X with maximal dimension. We Know that $\dim U \leq \dim X$ by 2.30. For the reverse inequality, we know that X can be covered by affine open sets. Since all of these sets are dense in X , they intersect pairwise, and therefore all have the same dimensions by 2.34 (since they have the same dimension as these intersections). Finally, the claim then follows by 2.34 (a). □

6 Chapter 6: Projective Varieties I: Topology

- 6.8 Let $R \neq 0$ be a graded ring. Show that the multiplicative unit $1 \in R$ is of degree 0.

Proof. Let d be the degree of 1 and $1 = \sum_{n \in \mathbb{N}} f_n$ be the homogeneous decomposition of 1. Since $1 = 1 \cdot 1$, $f_d \in R_d \cap R_{2d}$. Thus, since $f_d \neq 0$, we have $d = 0$ as desired. $\square$

- 6.14 Let $a \in \mathbb{P}^n$ be a point. Show that the one-point set $\{a\}$ is a projective variety, and compute explicit generators for $I_p(\{a\})$.

Proof. Let us write a in projective coordinates $a = (a_0 : \dots : a_n)$ where not all $a_i = 0$. Without loss of generality, let $a_0 \neq 0$ (otherwise just rename the variables). We claim that

$$I_p(\{a\}) = \langle x_1 - \frac{a_1}{a_0}x_0, \dots, x_n - \frac{a_n}{a_0}x_0 \rangle.$$

These are all homogeneous and a point in affine $(n+1)$ space vanishes on this set if and only if it is collinear to $(a_0, \dots, a_n)$. All the nonzero points of the line defined in this set are the same point in projective space. $\square$

- 6.29. Let $L_1, L_2 \subset \mathbb{P}^3$ be two disjoint lines (i.e. 1-dimensional linear subspaces in the sense of Example 6.13 (b)), and let $a \in \mathbb{P}^3 \setminus (L_1 \cup L_2)$. Show that there is a unique line $L \subset \mathbb{P}^3$ through a that intersects both L_1 and L_2 . Is the corresponding statement for lines and points in $\mathbb{A}^3$ true as well?

Proof. Let us view this question in the representation $\mathbb{P}^3 = (\mathbb{A}^4 \setminus \{0\})/\sim$. In this representation, L_1, L_2 correspond to planes;

$$\begin{aligned} L_1 &= \text{span}_k\{v_1, v_2\} \\ L_2 &= \text{span}_k\{w_1, w_2\} \end{aligned}$$

Since L_1 and L_2 intersect trivially in $\mathbb{P}^3$, their intersection in $\mathbb{A}^4$ is the origin. This, in turn, implies that $\{v_1, v_2, w_1, w_2\}$ form a basis of $k^4 \simeq \mathbb{A}^4$ (this isomorphism is as k -vector spaces). The point $a \in \mathbb{P}^3$ corresponds to a line $A = \text{span}_k\{x\}$ through the origin in $\mathbb{A}^4$. Due to the fact that $\{v_1, v_2, w_1, w_2\}$ form a basis of k^4 , we have unique representation

$$x = \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 w_1 + \lambda_4 w_2.$$

The assumption that $a \in \mathbb{P}^3$ does not lie on either line, is equivalent to the fact that neither are, λ_1 and λ_2 both zero, nor are λ_3, λ_4 both zero. Consider now the plane given by

$$E: \text{span}_k\{x, \lambda_1 v_1 + \lambda_2 v_2\}.$$

This plane clearly contains A and intersects both L_1 and L_2 in a line. Thus, the projective line corresponding to the affine plane E has the desired properties. It remains to address uniqueness. We will do this in the projective setting. Suppose line H the three properties. If H is different

from L , it has different intersection points with the with L_1 and L_2 , since a projective line is uniquely determined by two points (an affine plane is determined by two lines) let us write:

$$\begin{aligned} L \cap L_1 &= P_1, \quad L \cap L_2 = P_2 \\ H \cap L_1 &= Q_1, \quad H \cap L_2 = Q_2 \end{aligned}$$

These points are obviously coplanar, since they lay on the plane spanned by the lines L and H through a . But then, L_1 and L_2 are coplanar as well. Since two projective lines in the plane always intersect, we arrive at a contradiction and $H = L$ as desired.

The statement is not true in the affine case. It is already false in $\mathbb{A}^2$: taking for L_1, L_2 , any two parallel lines and a arbitrary, there are infinitely many lines through a which intersect both lines. $\square$

- 6.30. (a) Prove that a graded ring R is an integral domain if and only if for all homogeneous elements $f, g \in R$ with $fg = 0$ we have $f = 0$ or $g = 0$.
 (b) Show that a projective variety X is irreducible if and only if its homogeneous coordinate ring $S(X)$ is an integral domain.

Proof. (a) If R is an integral domain, and $f, g \neq 0$ with $\deg f = d_1$ and $\deg g = g_1$, then $\deg fg = d_1 + d_2$. In particular, it is nonzero. Conversely, if $f, g \neq 0$ and the leading term of f is f_{d_1} and that of g is g_{d_2} , then the leading term of fg is $f_{d_1}g_{d_2} \neq 0$. Thus $fg \neq 0$.

- (b) Suppose X is irreducible, we seek to show that $S(X)$ is an integral domain. By (a) it suffices to show that $S(X)$ has no (nonzero) homogeneous zero divisors. Thus, suppose $f, g \in S(X)$ are homogeneous and $f \cdot g = 0$. We obtain $X = V(fg) = V(f) \cup V(g)$. By the irreducibility of X , X is contained in one of these two sets. Let us Assume $X \subseteq V(f)$. (in particular $\sqrt{(f)} \neq I_0$). Applying the projective Nullstellensatz, we find $f = 0$. Conversely, suppose $X = V(I) \cup V(J) = V(I \cap J)$. Hilberts Nullstellensatz implies:

$$(0) = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J} \supseteq \sqrt{I} \cdot \sqrt{J}$$

Since $S(X)$ is an integral domain, one of the two above ideals must be zero, which is equivalent to the claim. $\square$

- 6.31. In this exercise we want to show that an intersection of projective varieties is never empty unless one would expect it to be empty for dimensional reasons (e.g. the phenomenon of parallel non-intersecting lines in the plane does not occur in projective space). So let $X, Y \subset \mathbb{P}^n$ be non-empty projective varieties. Show:

- (a) The dimension of the cone $C(X) \subset \mathbb{A}^{n+1}$ is $\dim X + 1$.
- (b) If $\dim X + \dim Y \geq n$ then $X \cap Y \neq \emptyset$.

Proof. Let $d := \dim X$ and suppose we have the maximal chain:

$$Z : \emptyset \neq Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_d \subseteq X$$

of closed irreducible sets. Forming the cones we obtain

$$C(Z) : \emptyset \neq C(Z_0) \subsetneq C(Z_1) \subsetneq \cdots \subsetneq C(Z_d) \subseteq C(X)$$

if $Z = V_P(I)$ it is easy to see that $C(Z) = V(I)$. This shows that the above cones are closed and that cones of irreducible closed sets are closed. Since the cone of a point is a line, we may extend this to a chain of length $d + 1$, by adding a one point set to the bottom of this chain. Thus, we have shown: $\dim C(X) \geq d + 1$. For the reverse inequality, we know that $\dim C(X)$ is given by the transcendence degree (over k) of its coordinate ring, which is given by $S(X)$. Using Noether's normalization $S(X) = k[x_1, \dots, x_d][d_1, \dots, d_k]$. After a suitable change of coordinates $\mathbb{A}^n$, (which translates to a homeomorphism of $\mathbb{P}^n$) we may assume the x_i to be homogeneous and thus, we obtain a chain in X given by

$$V_P(x_1, \dots, x_d) \subsetneq V_P(x_1, \dots, x_{d-1}) \subsetneq \cdots \subsetneq V_P(x_1) \subsetneq X$$

This is a chain of length $d + 1$. However, we can only conclude $\dim X \geq d$ since $(x_1, \dots, x_d)$ could (and does) correspond to the irrelevant ideal, which makes $V_P(x_1, \dots, x_d) = \emptyset$.

- (b) Without loss of generality, let X and Y irreducible, since otherwise we may choose irreducibility components of maximal dimension. This implies, that their cones are irreducible as well and in particular, also pure-dimensional. By 5.23 (b), and part (a) of this task, we then have that

$$\dim(C(X) \cap C(Y)) \geq \dim C(X) + \dim C(Y) - n = \dim X + \dim Y + 2 - n \geq 2.$$

Thus, $C(X) \cap C(Y)$ contains a nonzero point P . By the construction of cones, any multiple of P also lies in the cone, and therefore $C(X) \cap C(Y)$ actually contains a line. This line corresponds to a projective point in the intersection $X \cap Y$. $\square$

7 Chapter 7: Projective Varieties II: Ringed Spaces

- 7.3. Check that Definition 7.1 (together with Proposition 7.2) is compatible with our earlier constructions in the following cases:

- (a) The prevariety $\mathbb{P}^1$ is the same as the one introduced in Example 5.5 (a).
- (b) If $X \subset \mathbb{P}^n$ is a projective variety then its structure sheaf as defined above is the same as the closed subprevariety structure of X in $\mathbb{P}^n$ as in Construction 5.12 (b).

Proof. (a) We have the following two representations of $\mathbb{P}^1$ as a prevariety:

- (i) $\mathbb{P}^1$ as the gluing of two copies of $\mathbb{A}^1$ along the open sets $\mathbb{A}^1 \setminus \{0\}$ via the isomorphism $x \mapsto \frac{1}{x}$. The ringed space structure is the ringed structure induced by gluing; a function is regular if both pullbacks are regular.
- (ii) $\mathbb{P}^1$ as the topological space $(\mathbb{A}^2 \setminus \{0\})/\sim$, where collinear points are identified. The ringed structure was given by functions which are locally representable by quotients of homogeneous polynomials in $k[x_0, x_1]$ of the same degree.

denote by X the first space and by Y the second space. Let us exhibit an isomorphism between these two constructions. let us denote in X , the two copies of $\mathbb{A}^1$ and U_1 and U_2 respectively and i_1, i_2 , the two respective isomorphisms of $\mathbb{A}^1$ onto this set. Consider the following map:

$$\Phi : X \rightarrow Y, x \mapsto \begin{cases} (i_1^{-1}(x) : 1) & \text{if } x \in U_1 \\ (1 : i_2^{-1}(x)) & \text{if } x \in U_2 \end{cases}$$

This map is well defined due to the way we chose our gluing map, and it is a morphism (of prevarieties) since it is a morphism when restricted to either U_i (on these sets this looks just like the usual inclusions of affine space into projective space). Let us construct an inverse morphism. For this Let $V_0, V_1 \subseteq Y$ be the two canonical affine spaces and let us define the maps

$$\begin{aligned} \Psi_1 : V_1 &\rightarrow Y (x_0; x_1) \mapsto i_1(x_0/x_1) \\ \Psi_2 : V_0 &\rightarrow Y (x_0; x_1) \mapsto i_2(x_1/x_0) \end{aligned}$$

As compositions of morphisms, they are each morphisms to X . Since they agree on their intersection, they define a global morphism to X . It is immediate to verify, that these maps are inverses of one another.

- (b) This follows from the fact that the elements of the coordinate ring of X are restrictions of those on $\mathbb{P}^n$ and these rings agree on homogeneous elements.

□

7.6. Show by example that the homogeneous coordinate ring of a projective variety is *not* invariant under isomorphisms, i.e. there exist isomorphic projective varieties X, Y such that $S(X) \neq S(Y)$.

Proof. From example 7.5 (d) in the notes, we have an isomorphism

$$\mathbb{P}^2 \supseteq X := V_p(xy - z^2) \cong \mathbb{P}^1.$$

We may compute

$$S(X) = \frac{k[x, y, z]}{xy - z^2}$$

$$S(\mathbb{P}^1) = k[x, y]$$

These two rings are not isomorphic, since the latter is a UFD, while the former is not: $xy = z^2$ $\square$

7.7. Let $m, n \in \mathbb{N}_{>0}$. Prove:

- (a) If $f : \mathbb{P}^n \rightarrow \mathbb{P}^m$ is a morphism and $X \subset \mathbb{P}^m$ a hypersurface, then every irreducible component of $f^{-1}(X)$ has dimension at least $n - 1$.
- (b) If $n > m$ then every morphism $f : \mathbb{P}^n \rightarrow \mathbb{P}^m$ is constant.
- (c) $\mathbb{P}^n \times \mathbb{P}^m$ is not isomorphic to $\mathbb{P}^{n+m}$.

Proof. (a) By 5.24 it suffices to find for every irreducible component of $f^{-1}(X)$ a dense open subset of this component which has dimension at least $n - 1$. If $f^{-1}(X)$ is empty, the claim is true, since there are no irreducible components. If this is not the case, the canonical affine's cover $f^{-1}(X)$, and so for each component, we may choose such an affine open subset, and show that the intersection has dimension $\geq n - 1$. f restricted to a canonical affine is a morphism, and thus $f^{-1}(X)$ is given by $V_a(f^*p)$ where f^*p is a regular function on this affine. Thus, it is given globally by a polynomial. Furthermore, it is non-constant, since p is non-constant. By Krull's principle ideal theorem, $V_p(f^*p)$ has dimension $\geq n - 1$.

(b) If $n = 1$, then $m = 0$ and the claim is clear, so assume this not to be the case. Furthermore, assume for sake of contradiction, that the map is not constant. Consider the $n + 1$ -linear subspaces of $\mathbb{P}^m$ given by.

$$V(x_i) \subseteq \mathbb{P}^m.$$

If $X_i = f^{-1}(V(x_i)) = \emptyset$ for one of the i , then f may be interpreted as a map to the affine space $\mathbb{A}^n$ instead. We will show that each coordinate of this map is constant. Indeed, f restricted to any line is constant 5.7 (c) and 7.3 (a). Thus, f is globally constant since for any two projective points we may choose a line through these points and use the fact that f is constant on this line to conclude that the two projective points have the same image under f . Thus we can now assume $X_i \neq \emptyset$ for all $i = 0, \dots, m$. By part (a), the irreducible components of X_i all have dimension $\geq n - 1$. Thus,

$$\dim X_0 + \dim X_1 \geq n - 1 + n - 1 = 2n - 2 \geq n$$

Thus, by 6.31 (b) $X_0 \cap X_1 \neq \emptyset$. Restricting to afines, we find from 5.23, that each irreducible component of $X_0 \cap X_1$ has dimension $\geq n - 1 + n - 1 - n = n - 2$. Consider now:

$$(X_0 \cap X_1) \cap X_2$$

Using the fact that non of these are empty, and once again applying 6.31 and 5.23, we have that it has dimension $\geq n - 3$. Repeating this process, we eventually conclude that

$$\dim\left(\bigcap_{i=0}^m X_i\right) \geq n - m - 1 \geq 0.$$

In particular, this intersection is nonempty. This, however is a contradiction since:

$$\bigcap_{i=0}^m X_i = \bigcap_{i=0}^m f^{-1}(V_p(x_i)) = f^{-1}\left(\bigcap_{i=0}^m V_p(x_i)\right) = f^{-1}(\emptyset).$$

- (c) As the product space, $\mathbb{P}^n \times \mathbb{P}^m$ comes with canonical projections to each factor. These are clearly non-constant, and thus the existence of an isomorphism $\mathbb{P}^n \times \mathbb{P}^m \cong \mathbb{P}^{n+m}$ contradict (b).

□

7.8. Show that every morphism $f : \mathbb{P}^n \rightarrow \mathbb{P}^m$ must be of the form

$$x \mapsto (f_0(x) : \cdots : f_m(x))$$

as in Lemma 7.4, i.e. with $f_0, \dots, f_m \in K[x_0, \dots, x_n]$ homogeneous polynomials of the same degree such that $V_p(f_0, \dots, f_m) = \emptyset$.

Proof. Let $U_0 \subseteq \mathbb{P}^n$ be the open set given by $\{x_0 \neq 0\}$. We know that $U_i \cong \mathbb{A}^n$. Let $k[x_1, \dots, x_n]$, be the coordinate ring of $\mathbb{A}^n$, and let us restrict f , to U_i . Consider this as a morphism:

$$f' : \mathbb{A}^n \rightarrow \mathbb{P}^m, x \mapsto (f \circ i_i)(x)$$

where $i_i : \mathbb{A}^n \rightarrow U_i$ is the canonical inclusion isomorphism. Denoting by W_i , the canonical affines of $\mathbb{P}^m$, consider $f'^{-1}(W_i)$. We claim it is affine. Indeed, $f^{-1}(W_i) = \mathbb{P}^n \setminus f^{-1}(V_p(x_0))$. By 7.7, $f^{-1}(V_p(x_0))$ has dimension $\geq n - 1$. If this set has dimension n , then W_i is empty. If it has dimension $n - 1$, then it is a hypersurface and thus given by the zero locus of a homogeneous polynomial $f^{-1}(V_p(x_0)) = V_p(P_i)$. But this implies:

$$f'^{-1}(W_i) = \mathbb{A}^n \cap f^{-1}(W_i) = \mathbb{A}^n \cap (\mathbb{P}^n \setminus V_p(P)) = \mathbb{A}^n \setminus V_a(P_i^d),$$

where $q_i := P_i^d$ is the dehomogenization of P_i at the variable x_i . And this is a basic (also called distinguished) open set, which we know to be affine.

Let us denote the canonical isomorphism of $W_i \rightarrow \mathbb{A}^m$ by j_i . Then, in these coordinates, This morphisms are written as

$$g_i : f'^{-1}(W_i) \rightarrow \mathbb{A}^m, x \mapsto j_i \circ f'(x)$$

We know what the coordinate ring of $f'^{-1}(W_i)$ is: it is the localization $k[x_1, \dots, x_n]_{P_i^d}$. As a morphism between affine varieties, the coordinates of represented globally as a quotients of the form $\frac{A_t}{(P_i^d)^{k_t}}$ polynomial A_t and some power k . For simplicity, let us focus on $U_0 \cap W_i$. Using the isomorphisms i_0 and j_i to translate this back to the projective setting, we find :

$$\begin{aligned} f|_{U_0 \cap W_i}(x_0 : \dots : x_n) &= j_i^{-1} \circ g \circ i_0^{-1}(x_0 : \dots : x_n) \\ &= j_i^{-1} \circ g(x_1/x_0, \dots, x_n/x_0) \\ &= j_i^{-1} \left(\frac{A_1(x_1/x_0, \dots, x_n/x_0)}{(P_i^d)^{k_1}(x_1/x_0, \dots, x_n/x_0)}, \dots, \frac{A_m(x_1/x_0, \dots, x_n/x_0)}{(P_i^d)^{k_m}(x_1/x_0, \dots, x_n/x_0)} \right) \end{aligned}$$

We may multiply all coordinates a large power of $x_0 P_i^d$ to obtain a representation through homogenous polynomials which clearly all have the same degree in order for the expression to be well defined. We can do the same construction for the other $U_i \cap W_j$ to also obtain global homogeneous polynomial expressions for f on these sets. This shows that $\mathbb{P}^n$, may be covered by open sets on which the claim is true. Consider the representation of f on the open set $U_0 \cap W_0$. On this set we have:

$$f = (f_0 : \dots : f_m), \text{ with } V_p(f_0, \dots, f_m) \cap (U_0 \cap W_0) = \emptyset$$

Clearly we can rescale the f_i such that their GCD is 1. On a different open set $U_i \cap W_j$, we also have a global representation:

$$f = (g_0 : \dots : g_m) \text{ with } V_p(g_0, \dots, g_m) \cap (U_i \cap W_j) = \emptyset.$$

and we also may take these to have no GCD. On the intersection of their domains, the two representations agree:

$$(f_0 : \dots : f_n) = (g_0 : \dots : g_n) \Leftrightarrow \frac{f_i}{f_j} = \frac{g_i}{g_j} \text{ for all } i, j \Leftrightarrow f_i g_j = g_i f_j \text{ for all } i, j$$

The last equality is an equality of homogeneous polynomials. Therefore, the set where this equality holds is a closed set in $\mathbb{P}^n$. Since the equation holds on a nonempty open subset of $\mathbb{P}^n$, it holds globally. We claim that f_i and g_i have the same irreducible factors. Indeed, suppose F is a irreducible factor of f_i . Then The equality $f_i g_j = f_j g_i$ implies that F either divides f_j or g_i for all j . If F does not divide g_i , then F divides all f_j , which contradicts our assumption that their GCD is 1. Thus, F must divide g_i . In The same way the irreducible factors of g_i , divide f_i . In particular, $V_p(f_i) = V_p(g_i)$ for all i . However, this now implies:

$$V_p(f_0, \dots, f_m) \cap (U_i \cap W_j) = V_p(g_0, \dots, g_m) \cap (U_i \cap W_j) = \emptyset.$$

Since the sets $U_i \cap W_j$, cover $\mathbb{P}^n$, we obtain

$$V_p(f_0, \dots, f_m) = \bigcup_{i,j} (V_p(f_0, \dots, f_m) \cap (U_i \cap W_j)) = \emptyset.$$

Thus, we may choose $(f_1 : \dots : f_m)$ as a global representation of f . $\square$

- 7.14. Let $X \subset \mathbb{P}^2$ be a cubic curve. Moreover, let $U \subset X \times X$ be the set of all $(a, b) \in X \times X$ such that $a \neq b$ and the unique line through the two points a and b meets X in exactly three distinct points. Of course, two of these points must then be a and b ; we will denote the third one by $f(a, b) \in X$. Show that $U \subset X \times X$ is open, and that $f : U \rightarrow X$ is a morphism.

Proof. Let us first verify that there is indeed always a third intersection point. Suppose we have an arbitrary line $L \subseteq \mathbb{P}^2$. We have:

$$X = V_p(aX^3 + bY^3 + 3Z^2), \quad L = V_p(dX + eY + fZ),$$

where on each equation, not all coefficients are zero. Let us assume, wlog, that $d \neq 0$. by rescaling, we can assume $L = V_p(X + eY + fZ)$. Let us solve the system:

$$\begin{aligned} aX^3 + bY^3 + cZ^3 &= 0 \\ X + eY + fZ &= 0 \end{aligned}$$

Plugging the second condition, into the first:

$$-a(eY + fZ)^3 + bY^3 + cZ^3 = 0$$

If this is the zero equation, the L is a component of X , which is not a case we want to consider. For the same reason, the coefficients of Y^3 and Z^3 must be nonzero, since otherwise, both either X and Y divides, the equation, and we have infinity many solutions, so once again L would be a component of X . If the coefficient of Y^3 is nonzero, then dehomogenizing at Z , yields a cubic over k in one variable, and thus it has exactly three solutions, counted with multiplicity. The same holds, if instead the coefficient of Z^3 is nonzero. Let us assume wlog, the first case. We claim that the only solutions to our system lie at $Z \neq 0$. Indeed, if $Z = 0$, then we see that $Y = 0$ as well by considering the above cubic equation in Y and Z . Considering the equation for L , and recalling $a \neq 0$, we conclude that $X = 0$ as well. However, $V_p(X, Y, Z) = \emptyset$. Thus, for an irreducible cubic, the equation always has three solutions, when counted with multiplicity. Since a, b where two distinct points, there is a well defined line through them, which by the above discussion, intersects X a third time (when multiplicities are counted).

Next, we outline why the described map is actually a morphism. Although we do not give a full proof, the details tedious but not hard. It suffices to

prove that it is a morphism when restricted to affines. On these affines, the third point is given by solving a single linear equation whose coefficients are linear equations in the coordinates of a and b (a cubic on one variable divided by a quadratic). The solution of this linear equation is a morphism in the coefficients of the linear polynomial. $\square$

- 7.15. Show by example that not every hypersurface Y in a projective variety X is of the form $V(f)$ for a homogeneous polynomial $f \in S(X)$.

Proof. Consider the projective variety X given by $X = V_p(x_1x_4 - x_2x_3) \subseteq \mathbb{P}^3$. From task 2.40, we know that $V_X(x_1, x_2) \subseteq X$ is a projective subvariety of codimension 1 which is not the zero locus of a single polynomial (let alone homogeneous). $\square$

- 7.28. (a) For any $n, d \in \mathbb{N}_{>0}$, find explicit equations describing the image of the degree- d Veronese embedding $v_d : \mathbb{P}^n \hookrightarrow \mathbb{P}^N$, where $N = \binom{n+d}{n} - 1$.
(b) Prove that every projective variety is isomorphic to the zero locus of *quadratic* polynomials in a projective space.
(a) .

Proof. Let us denote the coordinates of $\mathbb{P}^N$ as :

$$Z_{i_0, \dots, i_n} \text{ where } \sum_{j=0}^n i_j = d.$$

we consider the Veronese embedding corresponding (in the obvious way) to this labeling of coordinates.

We may notice that whenever $(i_0, \dots, i_n) + (\tilde{i}_0, \dots, \tilde{i}_n) = (j_0, \dots, j_n) + (\tilde{j}_0, \dots, \tilde{j}_n)$, the points in the Veronese embedding satisfy:

$$Z_{i_0, \dots, i_n} \cdot Z_{\tilde{i}_0, \dots, \tilde{i}_n} = Z_{j_0, \dots, j_n} \cdot Z_{\tilde{j}_0, \dots, \tilde{j}_n}.$$

We claim that every point in $\mathbb{P}^N$, which satisfies the above equations, lies in the Veronese of $\mathbb{P}^n$. First, we claim that if a point $a \in \mathbb{P}^N$ satisfies the above equations then not all of its coordinates of the form $Z_i = Z_{0, \dots, d, \dots, 0}$ (which correspond to the tuple in which every entry except the i -th one is zero) can be zero. Indeed, suppose all such coordinates were in fact zero. Indeed, let $Z_{i,j}$ be the coordinate which corresponds to the $n+1$ -tuple which has $d-1$ in place i and 1 in place j . The governing equations yield:

$$(Z_{i,j})^2 = Z_i \cdot Z_{i,j}^{d-2,2} = 0$$

where $Z_{i,j}^{d-2,2}$ is the coordinate corresponding to the tuple which has entry $d-2$ in place i and entry 2 in place j . Using this iteratively,

we find that all coordinates corresponding to tuples with only two nonzero coordinates are zero. Following this procedure iteratively, we find in the next step that all coordinates corresponding to tuples with only three nonzero entry's are zero as well, and so on. Eventually, we conclude that all of a 's coordinates are zero, which is not possible as $a \in \mathbb{P}^N$.

Suppose now, that $a \in \mathbb{P}^N$ satisfies the given equations. We have shown that not all the coordinates of the form Z_i are zero, let us assume wlog, that for a , we have $z_0 \neq 0$. Consider the point $P \in \mathbb{P}^n$ given by

$$P = (z_0 : z_{0,1} : \dots : z_{0,d}) = \left(1 : \frac{z_{0,1}}{z_0} : \dots : \frac{z_{0,d}}{z_0}\right).$$

We claim that this point maps to a . Indeed, it agrees with a on the basic coordinates Z_i , and a point in the Veronese is uniquely determined by these coordinates as we have seen. $\square$

(b) The above equations are quadratic equations.

7.30. Recall from Example 7.5 (d) that a conic in $\mathbb{P}^2$ over a field of characteristic $\neq 2$ is an irreducible quadric curve.

- (a) Using the coefficients of quadratic polynomials, show that the set of all conics in $\mathbb{P}^2$ can be identified with an open subset U of the projective space $\mathbb{P}^5$. (One says that U is a *moduli space* for conics.)
- (b) Given a point $a \in \mathbb{P}^2$, show that the subset of U consisting of all conics passing through a is the zero locus of a linear equation in the homogeneous coordinates of $U \subset \mathbb{P}^5$.
- (c) Given 5 points in $\mathbb{P}^2$, no three of which lie on a line, show that there is a unique conic passing through all these points.
- (a) *Proof.* a quadric in $\mathbb{P}^2$ is given by the zero locus of equation $F(X, Y, Z) = 0$, where F is of the form

$$F(X, Y, Z) = aX^2 + bY^2 + cZ^2 + dXY + eXZ + fYZ.$$

where not all coefficients are zero. This polynomial is unique up to a nonzero scalar. Thus, we may assign to it a unique projective point $P = [a : b : c : d : e : f] \in \mathbb{P}^5$. Conversely we may assign to each point in $\mathbb{P}^5$ a quadratic equation, however not all such equations are irreducible. Nonetheless, this establishes a bijection between homogeneous quadratic equations in $k[X, Y, Z]$ and projective points in $\mathbb{P}^5$. We need to show that under this bijection, the irreducible homogeneous quadratic equations in $k[X, Y, Z]$ correspond to an open subset of $\mathbb{P}^5$. A homogeneous quadratic equation in $k[X, Y, Z]$ is reducible if and only if it factors as a product of (homogeneous) linear equations.

That is, $V(F)$ is a union of two (projective) lines in $\mathbb{P}^2$. Since two such lines always intersect, we see that there exists a projective point P , where we have:

$$F(P) = F_X(P) = F_Y(P) = F_Z(P) = 0.$$

(use the product rule at the intersection point of the lines). This is a homogeneous algebraic condition in the variables (a, b, c, d, e, f) and thus corresponds to a closed condition in $\mathbb{P}^5$. $\square$

(b) .

Proof. Let us write $a = [a_0 : a_1 : a_2]$. The conic corresponding to the projective point $P = [a : b : c : d : e : f]$ passes through this point if and only if:

$$aa_0^2 + ba_1^2 + ca_2^2 + da_0a_1 + ea_0a_2 + fa_1a_2 = 0.$$

This is a linear homogeneous equation in the coordinates of P . $\square$

- (c) For each of the 5 points, part (b) showed that the set of conics through these points correspond to the zero locus of linear equations in $U \subseteq \mathbb{P}^5$. Linear algebra shows, that the solution space of 5 linear equations in 6 variables is at least 1-dimensional. Thus, there is at least one quadratic equation, whose zero locus passes through the 5 points. This quadratic equation must be irreducible. If it were not, it would correspond to a line pair and the pigeon-hole principle ensures that there are 3 of the 5 points, which lay on one of these two lines. It remains to show uniqueness. Let $L_1, \dots, L_5$ be the five linear equations. We want to show that they are linearly independent, since in such a case there solution space would be 1-dimensional, hence corresponding to a unique projective point. Indeed, Since no three of the points are collinear, for each of the 5 points we may construct a conic which passes through all of the points except this point as follows: Let P_i be one of the five points and split the remaining four points into two pairs. Consider the pair of lines determined by these pairs. Since P lies on neither of these lines we have constructed the conic Q_i with:

$$Q_i(P_j) = 1 - \delta_{i,j}.$$

Assume now we have

$$\sum_{i=1}^d \lambda_i L_i = 0.$$

Plugging in the point Q_i shows that $\lambda_i = 0$ as desired.

7.31. Let $X \subset \mathbb{P}^3$ be the degree-3 Veronese embedding of $\mathbb{P}^1$, i.e., the image of

$$\mathbb{P}^1 \longrightarrow \mathbb{P}^3, \quad (x_0 : x_1) \longmapsto (y_0 : y_1 : y_2 : y_3) = (x_0^3 : x_0^2 x_1 : x_0 x_1^2 : x_1^3).$$

Moreover, let $a = (0 : 0 : 1 : 0) \in \mathbb{P}^3$ and $L = V(y_2) \subset \mathbb{P}^3$, and consider the projection f from a to L as in Example 7.5 (b).

- (a) Determine an equation of the curve $f(X)$ in $L \cong \mathbb{P}^2$.
(b) Is $f : X \rightarrow f(X)$ an isomorphism onto its image?

(a) *Proof.* The projection at a is given by the map

$$f: \mathbb{P}^3 \setminus \{a\} \rightarrow L, (y_0 : y_1 : y_2 : y_3) \mapsto (x_0 : x_1 : 0 : x_3).$$

Using the obvious isomorphism $L \simeq \mathbb{P}^2$, the task is asking about the image of the map. We may also notice that a is not in the image of the Veronese.

$$f: \mathbb{P}^1 \rightarrow \mathbb{P}^2, (x_0 : x_1) \mapsto (z_0 : z_1 : z_2) = (x_0^3 : x_0^2 x_1 : x_1^3).$$

We may notice that, the above points all satisfy the equation

$$z_1^3 - z_0^2 z_2 = 0.$$

This is a irreducible homogeneous equation (irreducibility follows since $z_0^2 z_2$ is not a cube in $k[z_0, z_2]$). Furthermore, Krull's principle ideal theorem ensures, that $V(z_1^3 - z_0^2 z_2)$ has projective dimension 1. Since $\mathbb{P}^1$ is complete and irreducible, the map f is either constant or surjective. Since $f(0 : 1) \neq f(1 : 0)$, the latter is true. Thus $f(X) = V(z_1^3 - z_0^2 z_2)$. $\square$

- (b) f is not an isomorphism. If it were, then it would restrict to an isomorphism when restricted to $\mathbb{P}^1 \setminus \{(1 : 0)\} \simeq \mathbb{A}^1$. The image of this set is $f(X) \setminus \{(1 : 0 : 0)\} = f(X) \cap U$, where $U = \{(z_0 : z_1 : z_2) | z_2 \neq 0\}$. Thus, in these coordinates we may interpret f as a map:

$$f: \mathbb{A}^1 \rightarrow V_a(z_1^3 - z_0^2) \subseteq \mathbb{A}^2, x \mapsto (t_1, t_2) = (x^3, x^2).$$

Although this map is a bijection, it is not an isomorphism since the induced map on coordinate rings is not surjective and thus not an isomorphism.